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# When Shift Happens - Confounding Is to Blame

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## Abstract

Distribution shift exposes a central limitation of contemporary machine learning: predictors that perform well on their training distribution can fail when deployed in environments generated by different underlying mechanisms. Although much of the OOD generalization literature emphasizes causal-invariant representations, recent empirical studies reveal a consequential and recurring pattern: (i) empirical risk minimization (ERM) can match or exceed specialized OOD generalization methods, and (ii) ERM often improves when it uses all available covariates rather than only variables designated as causal. This paper identifies hidden confounding as a principled explanation for that pattern. Through theoretical analysis and empirical evidence, we show that shifts in hidden confounders induce coupled changes in observable distributions and violate assumptions used by many existing OOD objectives. Under such shifts, predictive information is maximized not by discarding environment-specific structure, but by retaining environment-specific relationships together with invariant information. We further prove that covariates informative of hidden confounders can improve generalization by increasing conditional informativeness and reducing concept shift. These results recast the empirical success of all-variable ERM under real-world tabular shifts as a coherent information-theoretic phenomenon and identify concrete targets for OOD algorithms and covariate-selection strategies in confounded environments.

## 1 Introduction

Generalization—the capacity to draw reliable conclusions about unseen data from observed data—is a foundational objective across scientific and engineering disciplines. In medicine and the social sciences, it appears as *external validity*, which asks whether findings obtained in one population remain applicable in another [13]; in ecology, it motivates *space-for-time substitutions*, where spatial heterogeneity is used to infer temporal ecological patterns [60]; and in engineering, it underlies *robust control*, where systems must maintain performance despite unmodeled disturbances [35]. Machine learning now provides a central methodological framework for this broader problem, but its reliability depends on an explicit account of when learned relationships persist across environments [100, 88, 49].

For machine learning models, this requirement becomes robustness to distributional differences between training and deployment. A predictor trained to infer *food stamp reciprocity* from *household attributes* in one state, for example, should remain reliable when evaluated in another state with a different population and policy environment. We study this requirement as *out-of-distribution (OOD) generalization*: the predictor should maintain useful performance on in-distribution (ID) data sampled from the training distribution and on OOD test data drawn from distinct environments. Substantial work in domain generalization [56, 31, 53, 99, 6, 38, 74], domain adaptation [98, 82, 51, 90, 91, 81], and robust learning [42, 69, 94] provides guarantees under explicit structural or distributional assumptions. The empirical record, however, requires a sharper explanation. With careful model selection [30, 38, 87], standard *empirical risk minimization (ERM)* [86] often achieves competitive OOD performance in real-world applications [30, 24, 50, 57, 67, 87]. A study of 16 real-world tabular datasets [57] further reports that models trained on all available covariates can outperform models restricted to variables categorized as causal. These findings are not merely an implementation

anomaly: they indicate that generalization in real-world tabular data is not fully explained by an invariance-centered account alone.

Distribution shifts arise naturally when observations are collected from multiple environments [71]. In the food-stamp example, state-specific eligibility rules and population characteristics may jointly alter household attributes and reciprocity, thereby producing environment-indexed data-generating distributions. The dominant response in OOD learning is to seek *invariant relationships*: associations between covariates and outcomes that remain stable across environments [6]. Causal invariance is a particularly influential instance of this principle, because causal mechanisms are often expected to persist across interventions or domains [71, 59, 6]; consequently, many OOD methods attempt to identify representations or predictors that exploit stable causal or statistical relationships [6, 59, 31, 66, 56, 98, 63]. This strategy becomes incomplete when relevant variables are unobserved. Data-collection limits, privacy constraints, and measurement error routinely prevent all predictive variables from being recorded [14, 52, 20, 48]. When the missing variables confound observed covariates and the outcome, shifts in their distributions change the observable input–label relationship across environments. Such hidden confounding shifts are pervasive in realistic tabular settings; ignoring them can degrade generalization [40, 4, 85, 62] and can make empirically learned covariate–outcome relationships systematically misleading.

Despite its centrality to real-world generalization, hidden confounding shift remains under-characterized in the OOD literature [4, 40, 85, 62]. This paper develops a unified information-theoretic account of how hidden confounding reshapes predictive information, why environment-specific relationships can be necessary, and why non-causal but informative variables can improve performance. The analysis gives a principled explanation for empirical findings reported across recent OOD benchmarks [30, 24, 50, 57, 67, 87]. The paper makes four contributions:

- We formalize hidden confounding shift as a causal source of coupled observable distribution shifts and use this perspective to explain why informative, non-causal covariates can improve OOD performance (§ 3).
- We decompose predictive information between model predictions and outcomes and show that, under hidden confounding shift, maximizing this quantity requires environment-specific relationships in addition to invariant structure (§ 4.1, § 4.2).
- We prove that variables informative of the outcome or hidden confounders can improve predictive information by increasing conditional informativeness and reducing concept shift, thereby giving a principled account of covariate selection under hidden confounding shift (§ 4.3).
- Across real-world and synthetic datasets, we provide empirical evidence that (i) hidden confounding signatures are widespread in tabular benchmarks, (ii) environment-specific information terms align with ID and OOD test accuracy, and (iii) informative, non-causal covariates improve generalization (§ 5).

Taken together, these contributions recast the empirical success of all-variable ERM as an interpretable consequence of hidden confounding shift and identify information-theoretic targets for OOD methods that operate in confounded environments.

## 2 Related work

OOD generalization intersects several problem formulations, including domain generalization [56, 31, 53, 99, 6, 38], domain adaptation [98, 82, 51, 90, 91, 81], robust learning [42, 69, 94], federated learning [54, 36, 44, 97, 45], and OOD detection [41, 32, 92, 84, 49]. These settings differ in the information available to the learner: domain adaptation uses unlabeled test-domain data, domain generalization uses environment labels during training, federated learning addresses collaborative learning under communication and privacy constraints, and OOD detection identifies samples that depart from the training distribution. Across these formulations, invariance is a recurring design principle. Recent work establishes, however, that invariance alone is not sufficient to determine generalization quality [46, 72, 93]. In particular, Ye et al. [93] argue that invariant features must also remain informative. Our analysis places this invariance–informativeness trade-off within the specific and practically important regime of hidden confounding shift.

**Proxy variable adjustment:** When confounders are observed during training but absent at test time, Landeiro and Culotta [40] adjust for confounding shifts to improve classifier performance.

Related proxy-adjustment methods for causal effect estimation [55, 39] motivate recent domain-adaptation approaches that cast unknown distribution shift as a causal identification problem [4, 85]. OOD generalization under hidden confounding shift has also been analyzed under overlapping confounder support [62]. Our contribution is complementary and more diagnostic: rather than treating proxies only as an adjustment device, we characterize *how* proxy variables modify predictive-information terms and thereby improve OOD generalization.

**All variable models vs causal models.** Recently, Nastl and Hardt [57] introduced a benchmark in which covariates are grouped as causal, arguably causal, anti-causal, or otherwise spurious. Across 16 datasets, models trained on all covariates Pareto-dominate models restricted to causal or arguably causal subsets on both ID and OOD data, yet the theoretical basis for this pattern remains limited. We address this gap by showing why non-causal variables can be valuable when they are informative about hidden confounders or outcomes. For linear causal models, anchor regression [68] already frames estimation as a trade-off between using all observed covariates and privileging causal covariates. Our analysis specifies that trade-off under hidden confounding shift. Relatedly, Eastwood et al. [21] show that unstable covariates can improve performance when they carry label information and are conditionally independent of stable covariates given the label; their method adjusts distribution shift using unlabeled test-domain data. Nevertheless, ERM remains competitive on a medical real-world dataset not constructed for that setting [8], consistent with Nastl and Hardt [57]. This pattern also accords with the result that maximum likelihood estimation (MLE) is minimax optimal for OOD generalization under well-specified covariate shifts [26]. The present paper focuses on the harder and common case in which hidden confounding prevents such well specification.

### 3 Manifestations of hidden confounding shift

This section formalizes hidden confounding shift and explains why it is consequential for causal effect identification and prediction with observed covariates and outcomes.

**Types of distribution shifts.** For covariates  $\mathbf{X}$  and target  $Y$ , several distinct shifts can be observed between environments  $e$  and  $e'$ , as summarized in Table 1. Such shifts frequently arise from a change in the distribution  $\mathbb{P}(U)$  of an unobserved covariate  $U$  that causes  $\mathbf{X}$ ,  $Y$ , or both (Figure 1). A shift in  $\mathbb{P}(U)$  is then expressed indirectly through observed distributions involving only  $\mathbf{X}$  and  $Y$ . For example,  $\mathbb{P}(\mathbf{X}) = \sum_U \mathbb{P}(U)\mathbb{P}(\mathbf{X} | U)$ , so when  $\mathbb{P}^e(U) \neq \mathbb{P}^{e'}(U)$  and  $U \rightarrow \mathbf{X}$ , the observed covariate distribution changes as  $\mathbb{P}^e(\mathbf{X}) \neq \mathbb{P}^{e'}(\mathbf{X})$ . The same reasoning applies to shifts in  $\mathbb{P}(Y)$ ,  $\mathbb{P}(\mathbf{X} | Y)$ ,  $\mathbb{P}(Y | \mathbf{X})$ ,  $\mathbb{P}(\mathbf{X}, Y)$ .

Table 1: Canonical environment-level distribution shifts and the corresponding distributional inequalities.

| Type of Shift         | Mathematical Expression                                             |
|-----------------------|---------------------------------------------------------------------|
| Label                 | $\mathbb{P}^e(Y) \neq \mathbb{P}^{e'}(Y)$                           |
| Covariate             | $\mathbb{P}^e(\mathbf{X}) \neq \mathbb{P}^{e'}(\mathbf{X})$         |
| Conditional Covariate | $\mathbb{P}^e(\mathbf{X}   Y) \neq \mathbb{P}^{e'}(\mathbf{X}   Y)$ |
| Concept               | $\mathbb{P}^e(Y   \mathbf{X}) \neq \mathbb{P}^{e'}(Y   \mathbf{X})$ |
| Dataset               | $\mathbb{P}^e(\mathbf{X}, Y) \neq \mathbb{P}^{e'}(\mathbf{X}, Y)$   |

Several existing methods for OOD generalization assume certain causal relationships among  $U$ ,  $\mathbf{X}$ ,  $Y$  that guarantee specific invariances. For instance, when  $U \rightarrow Y \rightarrow \mathbf{X}$  (Figure 1 (a)), since  $\mathbb{P}(\mathbf{X}, Y) = \mathbb{P}(Y)\mathbb{P}(\mathbf{X} | Y)$ , a shift  $\mathbb{P}^e(\mathbf{X}, Y) \neq \mathbb{P}^{e'}(\mathbf{X}, Y)$  is observed due to label shift, i.e.,  $\mathbb{P}^e(Y) \neq \mathbb{P}^{e'}(Y)$ , but conditional

covariate distribution stays invariant i.e.,  $\mathbb{P}^e(\mathbf{X} | Y) = \mathbb{P}^{e'}(\mathbf{X} | Y)$  because  $\mathbb{P}(\mathbf{X}|Y, U) = \mathbb{P}(\mathbf{X}|Y)$  [89, 83, 25, 5, 47, 96, 18, 79]. Similarly, when  $U \rightarrow \mathbf{X} \rightarrow Y$  (Figure 1 (b)), a shift  $\mathbb{P}^e(\mathbf{X}, Y) \neq \mathbb{P}^{e'}(\mathbf{X}, Y)$  is observed due to covariate shift, i.e.,  $\mathbb{P}^e(\mathbf{X}) \neq \mathbb{P}^{e'}(\mathbf{X})$ , but the conditional distribution  $\mathbb{P}(Y | \mathbf{X})$  stays invariant i.e.,  $\mathbb{P}^e(Y | \mathbf{X}) = \mathbb{P}^{e'}(Y | \mathbf{X})$  [70, 16, 80, 11, 28]. These invariances may not hold in many scenarios because  $U$  usually causes both  $\mathbf{X}$  and  $Y$  (Figure 1 (c)) [50, 4, 40, 85, 62, 65, 64]. In this case, when  $\mathbb{P}(U)$  shifts between environments, label shift, covariate shift, conditional covariate shift, and concept shift can all occur simultaneously. Nevertheless, confounding shift induces an invariance:  $\mathbb{P}^e(\mathbf{X}, Y | U) = \mathbb{P}^{e'}(\mathbf{X}, Y | U)$  referred to as *stable confounding shift* [85, 4]. Since  $U$  is unobserved, this invariance is absent from the observed data.

**Information-theoretic measures:** Following Federici et al. [22], we use mutual information to quantify distribution shifts. Specifically, we use  $I(\mathbf{X}; E)$ ,  $I(Y; E)$ ,  $I(\mathbf{X}; E|Y)$ ,  $I(Y; E|\mathbf{X})$ ,  $I(\mathbf{X}, Y; E)$

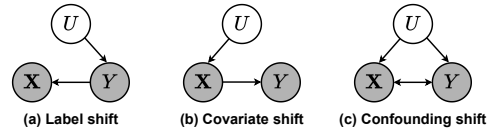


Figure 1: Causal graphs underlying distribution shifts.

### Structural Equations :

(Hidden Confounder)  $U \sim \mathcal{N}(\mu_e, 0.5^2)$   
 $\mu_e \in \{-2, 2\}$  for ID  
 $\mu_e \in \{0, 4\}$  for OOD  
(Input)  $X = 0.75U + \mathcal{N}(0, 1.5^2)$   
(Outcome)  $Y = X - 3U + \mathcal{N}(0, 0.5^2)$   
(Informative)  $X_i = 0.6U + \mathcal{N}(0, 0.3^2)$

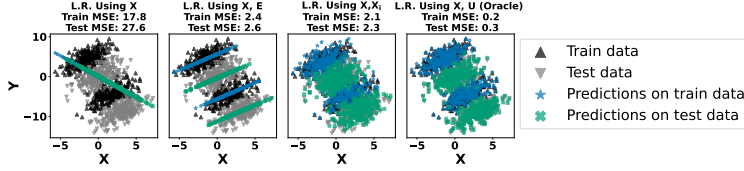


Figure 2: Four linear regression (L.R.) models evaluated in an OOD setting with hidden confounding shift and minimal environment overlap (i.e., distant  $\mu_e$ ). (i) A model trained only on  $X$  learns the wrong  $X$ – $Y$  relationship, illustrating Simpson’s paradox. (ii) Environment-specific summary statistics of  $X$ , denoted by  $E$ , recover the correct direction but remain representationally limited. (iii) An informative covariate  $X_i$  for  $U$  improves OOD generalization. (iv) The oracle model is trained on  $X$  and  $U$ .

where  $E$  is the environment variable to measure the shifts in  $\mathbb{P}(\mathbf{X})$ ,  $\mathbb{P}(Y)$ ,  $\mathbb{P}(\mathbf{X} \mid Y)$ ,  $\mathbb{P}(Y \mid \mathbf{X})$ ,  $\mathbb{P}(\mathbf{X}, Y)$  respectively across the environments. As discussed earlier, existing methods for OOD generalization often assume either  $I(\mathbf{X}; E \mid Y) = 0$  through the label shift assumption or  $I(Y; E \mid \mathbf{X}) = 0$  through the covariate shift assumption. However, as discussed earlier, both of them may be nonzero under hidden confounding shifts.

**Challenges with hidden confounding shift:** To illustrate the challenges posed by hidden confounding shift on OOD generalization, we present the experiment shown in Figure 2. The outcome  $Y$  depends on  $X$  and a hidden confounder  $U$ . Environment-specific shifts in the mean of  $U$ , denoted by  $\mu_e$ , induce distinct environments. A linear regression model trained solely on observed covariate  $X$  infers an incorrect relationship between  $X$  and  $Y$ , as shown in the first subplot. This exemplifies the Simpson’s paradox [73], where  $Y$  increases with  $X$  within each environment, yet the model captures the opposite trend. We observe similar behavior from the models designed for OOD generalization, such as IRM [6], VREX [38], and Group DRO [69]. In such scenarios, we theoretically show in § 4 that the optimal strategy for generalization involves learning environment-specific relationships, which enables recovery of the correct relationship between  $X$  and  $Y$ . This is illustrated in the second subplot, where environment-specific summary statistics of  $X$ , such as mean, standard deviation, and quantiles, help uncover the true  $X$ – $Y$  relationship. This mirrors the *backdoor adjustment* criteria in causal effect estimation [58], where environment-specific information acts as a proxy for adjusting for confounders. However, as shown in the second subplot, environment-specific information encoded in observed covariates alone may be insufficient due to the limited representational capacity. However, as we show in § 4.3, additional informative covariates serving as proxies for unobserved confounders can further improve generalization performance, as shown in the third subplot. For comparison, we include an oracle model trained on  $X$  and  $U$ , which achieves the best-possible mean squared error.

## 4 Impact of hidden confounding shifts on OOD generalization

This section develops the theoretical account of how hidden confounding shift alters the objectives and information quantities that govern OOD generalization. Let  $\hat{Y} = (f \circ \phi)(\mathbf{X})$  be the predicted label for an input  $\mathbf{X}$ , where  $f$  is a classifier and  $\phi$  is a feature extractor or transformation function.  $H(X) = -\mathbb{E}_X[\log(\mathbb{P}(X))]$  denotes entropy and  $I(X; Y) = \mathbb{E}_{X,Y}[\log \frac{\mathbb{P}(X,Y)}{\mathbb{P}(X)\mathbb{P}(Y)}]$  denotes mutual information. The risk of the predictor  $(f \circ \phi)$  in an environment  $e$  is defined as:  $\mathcal{R}^e(f \circ \phi) = \mathbb{E}_{(\mathbf{X}, Y) \sim \mathbb{P}_{\mathbf{X}, Y}^e}[\ell((f \circ \phi)(\mathbf{X}), Y)]$  where  $\ell(\cdot, \cdot)$  is a loss function. OOD generalization seeks a predictor  $(f \circ \phi)$  whose risk remains low on both ID and OOD data, and the literature has proposed several objective classes for this purpose. Robust optimization-based methods [7, 9, 19, 75, 69] aim to minimize the worst risk across all training environments:  $\mathcal{R}^{\text{Rob}}(f \circ \phi) = \max_{e \in \mathcal{E}_{tr}} \mathcal{R}^e(f \circ \phi)$  where  $\mathcal{E}_{tr}$  denotes the set of training environments. The variance risk extrapolation (VREX) [38] aims to minimize  $\mathcal{R}^{\text{VREX}}(f \circ \phi) = \sum_{e \in \mathcal{E}_{tr}} \mathcal{R}^e(f \circ \phi) + \beta \times \text{Variance}(\{\mathcal{R}^1(f \circ \phi), \dots, \mathcal{R}^{|\mathcal{E}_{tr}|}(f \circ \phi)\})$  where  $\beta$  is a hyperparameter. Invariant risk minimization (IRM) [6] aims to minimize  $\sum_{e \in \mathcal{E}_{tr}} \mathcal{R}^e(f \circ \phi)$  with the constraint that  $f$  is a simultaneously optimal classifier across all environments. Empirical risk minimization (ERM) methods pool observations from all training environments and minimize the empirical risk on the resulting sample [6, 38]. That is, ERM minimizes  $\mathcal{R}^{\text{ERM}}(f \circ \phi) = \sum_{e \in \mathcal{E}_{tr}} \sum_{(\mathbf{X}, Y) \in \mathcal{D}^e} \ell((f \circ \phi)(\mathbf{X}), Y)$  where  $\mathcal{D}^e$  denotes observed data in an environment  $e$ .

To analyze how hidden confounding shift affects these objectives, we study the predictive information  $I(Y; \hat{Y})$ , where  $\hat{Y} = (f \circ \phi)(\mathbf{X})$ . This mutual information quantifies how much the prediction  $\hat{Y}$  reveals about the true label  $Y$  and therefore provides a natural information-theoretic lens for

prediction. We begin by defining two properties of  $\phi(\mathbf{X})$ —*informativeness* and *invariance*—that jointly determine OOD generalization quality [93].

**Definition 4.1** (Informativeness and Conditional Informativeness). *The informativeness of features  $\phi(\mathbf{X})$  for predicting  $Y$  is defined as  $INF(\phi(\mathbf{X}), Y) = I(\phi(\mathbf{X}); Y)$ . The conditional informativeness of features  $\phi(\mathbf{X})$  for predicting  $Y$  conditioned on environment variable  $E$  is defined as  $CINF(\phi(\mathbf{X}), Y, E) = I(\phi(\mathbf{X}); Y|E)$ .*

Conditional informativeness measures the amount of label information retained by  $\phi(\mathbf{X})$  within environments. Minimizing CINF implies information loss, and maximizing CINF can be undesirable in applications such as algorithmic fairness, as  $\phi(\mathbf{X})$  may exploit sensitive or biased information within environments, leading to unfair predictions. While such biased representations can be avoided by minimizing  $I(\phi(\mathbf{X}); E)$ , it may reduce predictive performance, as we show in § 4.2. In the setting considered here, where  $E$  is not treated as sensitive information, we analyze the regime in which increasing CINF improves generalization, and we substantiate this perspective in § 4.1.

**Definition 4.2** (Variation and Invariance). *For a given label  $Y$ , the variation in the features  $\phi(\mathbf{X})$  across environments  $E$  is defined as  $VAR(\phi(\mathbf{X}), Y, E) = I(\phi(\mathbf{X}); E|Y)$ . For a given label  $Y$ , the invariance of the features  $\phi(\mathbf{X})$  across environments  $E$  is defined as the negative of the variation i.e.,  $INV(\phi(\mathbf{X}), Y, E) = -VAR(\phi(\mathbf{X}), Y, E) = -I(\phi(\mathbf{X}); E|Y)$ .*

Invariance quantifies the extent to which the representation  $\phi(\mathbf{X})$  remains stable across environments after conditioning on a fixed label  $Y$ . Minimizing invariance can lead to overfitting by preserving environment-specific information in  $\phi(\mathbf{X})$ , whereas maximizing invariance helps eliminate environment-specific dependencies, promoting invariant learning. Extending the information-theoretic measures from § 3, we use  $I(\phi(\mathbf{X}); E)$  and  $I(Y; E | \phi(\mathbf{X}))$  to quantify *feature shift* and *concept shift* respectively. Depending on the context, we use the term *concept shift* to denote either  $I(Y; E | \mathbf{X})$  or  $I(Y; E | \phi(\mathbf{X}))$ .

#### 4.1 A general decomposition of predictive information

We first state the causal-graph preliminaries used in the decomposition. A causal graph  $\mathcal{G}$  consists of nodes representing random variables, and directed edges indicating direct causal influences between nodes. A *path* between two nodes  $X_i$  and  $X_j$  is a sequence of unique nodes connected by edges. A *directed path* from  $X_i$  to  $X_j$  with  $i < j$  is one where all arrows point toward  $X_j$ , i.e.,  $X_i \rightarrow X_{i+1} \rightarrow \dots \rightarrow X_{j-1} \rightarrow X_j$ . In such a directed path,  $X_i$  is called the *parent* of  $X_{i+1}$ ,  $X_{i+1}$  is the *child* of  $X_i$ ,  $X_i$  an *ancestor* of  $X_j$ , and  $X_j$  is a *descendant* of  $X_i$ . Paths decompose into three fundamental structures: a *chain*  $X_i \rightarrow X_j \rightarrow X_k$ , a *fork*  $X_i \leftarrow X_j \leftarrow X_k$ , and a *collider*  $X_i \rightarrow X_j \leftarrow X_k$ . In both chains and forks,  $X_i$  and  $X_k$  are marginally dependent yet become conditionally independent upon conditioning on the intermediate node  $X_j$ , i.e.  $X_i \not\perp\!\!\!\perp X_k$  and  $X_i \perp\!\!\!\perp X_k | X_j$ . In a collider,  $X_i$  and  $X_k$  are marginally independent but become conditionally dependent when conditioning on  $X_j$  or any of its descendants, i.e.  $X_i \perp\!\!\!\perp X_k$  and  $X_i \not\perp\!\!\!\perp X_k | X_j$ . A path between  $X_i$  and  $X_k$  is said to be *blocked* by a conditioning set  $\mathcal{S}$  if and only if either (i) the path contains a chain or fork whose middle node lies in  $\mathcal{S}$ , or (ii) it contains a collider such that neither the collider nor any of its descendants belongs to  $\mathcal{S}$ . If all paths from  $X_i$  to  $X_k$  are blocked by  $\mathcal{S}$ , then  $X_i \perp\!\!\!\perp X_k | \mathcal{S}$ . A path is open if it is not blocked.

Now consider the predictive information  $I(Y; \hat{Y})$ , where the predictions  $\hat{Y} = (f \circ \phi)(\mathbf{X})$  are based on a learned representations  $\phi(\mathbf{X})$ . We model the underlying causal relationships among  $\mathbf{X}, Y, U, \phi(\mathbf{X}), \hat{Y}$  and  $E$  as shown in Figure 3. Here,  $U$  is a hidden confounding variable. Given either  $\mathbf{X}$  or  $\phi(\mathbf{X})$ ,  $\hat{Y}$  is redundant for reasoning about  $Y$ . That is,  $\hat{Y} \perp\!\!\!\perp Y | \mathbf{X}$  and  $\hat{Y} \perp\!\!\!\perp Y | \phi(\mathbf{X})$ .  $E$  denotes an environment variable that captures shifts in  $\mathbb{P}(U)$ . That is, between any two environments, there is a shift in  $\mathbb{P}(U)$ . We first present a more general decomposition of predictive information without explicit consideration of how  $U$  influences  $\mathbf{X}$  and  $Y$ . A formal treatment of  $U$ 's influence on  $\mathbf{X}$  and  $Y$  is presented in § 4.2.

**Proposition 4.1.** *For a covariate vector  $\mathbf{X}$ , label  $Y$ , with causal structure  $\mathbf{X} \leftrightarrow Y$ , i.e., some covariates cause  $Y$  and some covariates are caused by  $Y$ , environment variable  $E$ , a feature*

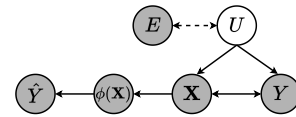


Figure 3: The bi-directed edge between  $\mathbf{X}$  and  $Y$  denotes the setting in which some covariates in  $\mathbf{X}$  may cause  $Y$ , while others may be caused by  $Y$ .

246 extractor  $\phi$ , and prediction  $\hat{Y}$ , the predictive information  $I(Y; \hat{Y})$  is decomposed as follows:

$$I(Y; \hat{Y}) = \overbrace{I(\phi(\mathbf{X}); Y|E)}^{\text{Cond. informativ.}} - \overbrace{\frac{I(\phi(\mathbf{X}); E|Y)}{2}}^{\text{Variation}} + \overbrace{\frac{I(Y; E)}{2}}^{\text{Label shift}} + \overbrace{\frac{I(\phi(\mathbf{X}); E)}{2}}^{\text{Feature shift}} - \overbrace{\frac{I(Y; E|\phi(\mathbf{X}))}{2}}^{\text{Concept shift}} - \overbrace{I(\phi(\mathbf{X}); Y|\hat{Y})}^{\text{Residual}} \quad (1)$$

247 where  $I(\phi(\mathbf{X}); Y|\hat{Y})$  is the residual information in  $\phi(\mathbf{X})$  for inferring  $Y$  that is not captured by the  
 248 prediction  $\hat{Y}$ . The decomposition above also holds when  $\phi(\mathbf{X})$  is replaced with  $\mathbf{X}$ .

249 Proofs are presented in Appendix § A. Existing methods can be viewed as methods that explicitly  
 250 minimize or maximize certain terms in Equation (1). For instance, IRM [6] enforces that a fixed  
 251 classifier  $f$  remains the same across different environments. This is directly related to maximizing  
 252  $-I(\phi(\mathbf{X}); E|Y)$  because if  $\phi(\mathbf{X})$  were to carry extra environment-specific information conditional  
 253 on  $Y$ , the fixed classifier would no longer be optimal in all environments. DANN [23] and the  
 254 independence criterion for fair classification [22] aims to minimize environment-specific information  
 255 in  $\phi(\mathbf{X})$  by minimizing  $I(\phi(\mathbf{X}); E)$ . In the context of domain adaptation, certain properties of  
 256 features are learned invariant to domains [82, 90]. While minimizing  $I(\phi(\mathbf{X}); E)$  may be good  
 257 for fair classification, it may degrade predictive performance [98, 33, 22]. This can be seen from  
 258 Equation (1), where minimizing  $I(\phi(\mathbf{X}); E)$  can reduce  $I(Y; \hat{Y})$ . In CDAN [51], the objective is  
 259 to obtain  $\mathbb{P}^e(\phi(\mathbf{X}), Y) = \mathbb{P}^e(\phi(\mathbf{X}), Y)$ . By enforcing this equality,  $E$  becomes independent of  
 260 the pair  $(\phi(\mathbf{X}), Y)$ . Consequently,  $\mathbb{P}(Y | \phi(\mathbf{X}), E) = \mathbb{P}(Y | \phi(\mathbf{X})) \implies I(Y; E | \phi(\mathbf{X})) = 0$ .  
 261 Thus, although CDAN does not explicitly include  $I(Y; E | \phi(\mathbf{X}))$  in its loss, its joint-distribution  
 262 alignment objective effectively drives this conditional mutual information to zero. While  $I(Y; E)$  is  
 263 a constant, there exist methods that are robust to shifts in the distribution of labels [69, 95].

264 Recall that Equation (1) is derived without explicitly considering how  $U$  influences  $\mathbf{X}, Y$ . Knowing  
 265 how  $U$  influences  $\mathbf{X}$  and  $Y$  can further guide the effective handling of the terms in (1). That is, if  
 266  $U \rightarrow Y, U \not\rightarrow \mathbf{X}, Y \rightarrow \mathbf{X}$ , then  $I(\mathbf{X}; E | Y) = 0$ , suggesting that minimizing  $I(\phi(\mathbf{X}); E | Y)$   
 267 is a principled objective. Similarly, if  $U \not\rightarrow Y, U \rightarrow \mathbf{X}, \mathbf{X} \rightarrow Y$ , then  $I(Y; E | \mathbf{X}) = 0$ , thereby  
 268 motivating the minimization of  $I(Y; E | \phi(\mathbf{X}))$ . However, if  $U \rightarrow Y$  and  $U \rightarrow \mathbf{X}$ , the interaction  
 269 between the terms in Equation (1) becomes non-trivial, and it remains unclear what constitutes an  
 270 ideal strategy for addressing them. We answer this question in the next section.

## 271 4.2 Predictive information under hidden confounding shift

272 To understand how hidden confounding variable  $U$  that cause both  $\mathbf{X}$  and  $Y$ , impact the predictive  
 273 information, we consider *two special cases* of the relationship between  $\mathbf{X}$  and  $Y$ : (i)  $\mathbf{X} \rightarrow Y$  and  
 274 (ii)  $Y \rightarrow \mathbf{X}$ . When  $\mathbf{X} \rightarrow Y$ , we obtain the inequalities in (2) because conditioning on  $Y$  opens the  
 275 path  $\phi(\mathbf{X}) \leftarrow \mathbf{X} \rightarrow Y \leftarrow U \leftarrow E$  from  $\phi(\mathbf{X})$  to  $E$  at the collider node  $Y$  which results in additional  
 276 information flow from  $\phi(\mathbf{X})$  to  $E$  via  $Y$ . In contrast, conditioning on  $\phi(\mathbf{X})$  partially blocks the  
 277 information flow from  $Y$  to  $E$  at the node  $\mathbf{X}$  as long as  $\phi(\mathbf{X})$  encodes some information about  $\mathbf{X}$ .

$$(i) \overbrace{I(\phi(\mathbf{X}); E|Y)}^{\text{Variation}} \geq \overbrace{I(\phi(\mathbf{X}); E)}^{\text{Feature shift}} \quad (ii) \overbrace{I(Y; E)}^{\text{Label shift}} \geq \overbrace{I(Y; E|\phi(\mathbf{X}))}^{\text{Concept shift}} \quad (2)$$

278 Similarly, when  $Y \rightarrow \mathbf{X}$ , we obtain the inequalities in (3) because conditioning on  $Y$  blocks  
 279 the information flow from  $\phi(\mathbf{X})$  to  $E$  at the node  $Y$  and conditioning on  $\phi(\mathbf{X})$  opens the path  
 280  $Y \rightarrow \mathbf{X} \leftarrow U \leftarrow E$  from  $Y$  to  $E$  at the node  $\mathbf{X}$  because  $\phi(\mathbf{X})$  is the child of  $\mathbf{X}$  and conditioning on  
 281 the child of a collider opens a path via that collider [58] (recall the preliminaries in § 4.1).

$$(i) \overbrace{I(\phi(\mathbf{X}); E)}^{\text{Feature shift}} \geq \overbrace{I(\phi(\mathbf{X}); E|Y)}^{\text{Variation}} \quad (ii) \overbrace{I(Y; E|\phi(\mathbf{X}))}^{\text{Concept shift}} \geq \overbrace{I(Y; E)}^{\text{Label shift}} \quad (3)$$

282 Using the inequalities in (2), (3), we refine the predictive information decomposition as follows.

283 **Proposition 4.2.** For a covariate vector  $\mathbf{X}$ , label  $Y$ , an environment variable  $E$ , a feature extractor  
 284  $\phi$ , the prediction  $\hat{Y}$ , and an unobserved confounding variable  $U$  that cause both  $\mathbf{X}$  and  $Y$ , and if  
 285 either (i)  $\mathbf{X} \rightarrow Y$  or (ii)  $Y \rightarrow \mathbf{X}$ , the predictive information  $I(Y; \hat{Y})$  can be decomposed as follows:

$$I(Y; \hat{Y}) = \overbrace{I(\phi(\mathbf{X}); Y|E)}^{\text{Cond. informativeness}} - \overbrace{I(\phi(\mathbf{X}); Y|\hat{Y})}^{\text{Residual}} \quad (4)$$

286 In Proposition 4.2, we tackle the two cases:  $\mathbf{X} \rightarrow Y$  and  $Y \rightarrow \mathbf{X}$  separately. However, the causal  
 287 relationship  $\mathbf{X} \rightarrow Y$  is more prevalent in real-world tabular prediction tasks. For instance, 11 out of  
 288 16 datasets considered in recent benchmark studies [57, 50, 24] follow the causal structure  $\mathbf{X} \rightarrow Y$ .  
 289 In the remaining 5 out of 16 datasets, the number of covariates that are caused by  $Y$  is much smaller  
 290 than the number of covariates that cause  $Y$ . We present the key takeaway from Proposition 4.2 below.

Compared to the decomposition in (1), the decomposition in (4) (which is obtained for the special cases: (i)  $U \rightarrow \mathbf{X}, U \rightarrow Y, \mathbf{X} \rightarrow Y$ , (ii)  $U \rightarrow \mathbf{X}, U \rightarrow Y, Y \rightarrow \mathbf{X}$ ) is free from the terms: *variation, label shift, feature shift, and concept shift*. That is, under hidden confounding shift, when either  $\mathbf{X} \rightarrow Y$  or  $Y \rightarrow \mathbf{X}$ , the predictive information can be maximized by maximizing the difference: *conditional informativeness* – *residual*.

291

292 **Maximizing conditional informativeness:** In a very recent work, Prashant et al. [62] proposed a  
 293 mixture of experts (MoE) model for OOD generalization under hidden confounding shift, where  
 294 each expert corresponds to a specific hidden confounder assignment. That is, each expert focuses on  
 295 maximizing the performance within the environment corresponding to a particular value of the hidden  
 296 confounder. Thus, learning MoE models aligns with the goal of maximizing the conditional infor-  
 297 mativeness  $I(\phi(\mathbf{X}); Y | E)$ . Equation (4) provides theoretical justification for the use of MoE-type  
 298 models. Beyond supporting such models, the Equation (4) also highlights the need for methods that  
 299 operate under more general confounding shift settings. For instance, the method proposed by Prashant  
 300 et al. [62] assumes overlapping confounding support i.e.,  $\text{supp}(\mathbb{P}(U | \mathcal{E}_{te})) \subseteq \text{supp}(\mathbb{P}(U | \mathcal{E}_{tr}))$ , and  
 301 the proxy variables to be discrete-valued.

The results in Figure 2 illustrate that, in a simple setting without confounding overlap and with a continuous-valued proxy, a linear regression model can successfully recover the correct causal relationship between observed covariates and the target across environments when provided with environment-specific information. This serves as a motivational example that emphasizes the need for OOD generalization methods that remain effective even when confounding support is limited and the confounder is continuous-valued.

302

303 **Minimizing residual:** The residual term  $I(\phi(\mathbf{X}); Y | \hat{Y})$  quantifies how much additional information  
 304  $\phi(\mathbf{X})$  provides about the true label  $Y$  beyond what is already contained in the prediction  $\hat{Y}$ . The  
 305 residual term can be expressed as:  $I(\phi(\mathbf{X}); Y | \hat{Y}) = H(Y | \hat{Y}) - H(Y | \phi(\mathbf{X}), \hat{Y})$ . The conditional  
 306 entropy  $H(Y | \hat{Y})$  is related to the expected cross-entropy loss as:  $\mathbb{E}_{Y, \hat{Y}}[\ell_{\text{CE}}(Y, \hat{Y})] = H(Y | \hat{Y})$   
 307  $+ \text{calibration error}$  [12, 10, 29]. Here  $H(Y | \hat{Y})$  is known as the *refinement error* that measures  
 308 the model’s ability to distinguish between classes. Minimizing the residual term  $I(\phi(\mathbf{X}); Y | \hat{Y})$  thus  
 309 involves two complementary strategies. First, one can directly minimize  $I(\phi(\mathbf{X}); Y | \hat{Y})$  by making  
 310  $\phi(\mathbf{X})$  conditionally independent of the target  $Y$  given  $\hat{Y}$ , i.e.,  $Y \perp\!\!\!\perp \phi(\mathbf{X}) | \hat{Y}$ . Alternatively, one can  
 311 minimize the refinement error  $H(Y | \hat{Y})$ , for instance, through training techniques that minimize  
 312 refinement error for accurate predictions before minimizing the calibration error [10].

### 313 4.3 Impact of additional informative variables

314 In § 4.2, we present the optimal strategies for generalization when the confounder is unobserved.  
 315 In practice, we sometimes have access to proxies for the hidden confounding variable [4, 85, 62],  
 316 which can be leveraged to substitute for the hidden confounding variable. We now study the impact  
 317 of adding such non-causal but informative covariates on predictive information.

318 **Definition 4.3** (Informative Covariates). *A set of covariates  $\mathbf{X}_I$  that are not causally related to  $Y$  i.e.,*  
 319 *neither ancestors nor descendants of  $Y$  are in  $\mathbf{X}_I$ , are said to be informative to  $Y$  if  $\mathbf{X}_I$  and  $Y$  are*  
 320 *not independent of each other given other causally related covariates  $\mathbf{X}$  and  $E$  i.e.,  $Y \not\perp\!\!\!\perp \mathbf{X}_I | \mathbf{X}, E$ .*

321 Since  $U \rightarrow Y$ , any covariate that is informative to  $U$ , is also informative to  $Y$  and vice-versa. From  
 322 the predictive information decomposition in (1), to maximize  $I(Y; \hat{Y})$ , it is required to minimize the  
 323 concept shift  $I(Y; E | \phi(\mathbf{X}))$ . Liu et al. [50] suggest based on their empirical analysis that collecting  
 324 additional covariates  $\mathbf{X}_I$  such that  $\mathbb{P}(Y | \mathbf{X}, \mathbf{X}_I)$  is more stable across environments i.e., reducing

Table 2: Quantifying distribution shifts. Mean $\pm$ standard deviation is computed over 10 random subsets of 40,000 samples. Statistical significance against a mean of zero is assessed via one-sample t-tests ( $\alpha = 0.05$ ), confirming all measures are significantly different from zero (p-value  $\approx 0$ ).

| Dataset         | Conditional Covariate Shift<br>$I(\mathbf{X}; E Y)$ | Label shift<br>$I(Y; E)$ | Covariate Shift<br>$I(\mathbf{X}; E)$ | Concept shift<br>$I(Y; E \mathbf{X})$ |
|-----------------|-----------------------------------------------------|--------------------------|---------------------------------------|---------------------------------------|
| Readmission     | $0.107 \pm 0.002$                                   | $0.068 \pm 0.002$        | $0.097 \pm 0.002$                     | $2.032 \pm 0.000$                     |
| Food stamps     | $0.126 \pm 0.004$                                   | $0.030 \pm 0.003$        | $0.108 \pm 0.001$                     | $2.118 \pm 0.001$                     |
| Income          | $0.168 \pm 0.002$                                   | $0.075 \pm 0.003$        | $0.147 \pm 0.001$                     | $2.059 \pm 0.002$                     |
| Public coverage | $0.231 \pm 0.002$                                   | $0.412 \pm 0.006$        | $0.222 \pm 0.002$                     | $1.945 \pm 0.001$                     |
| Unemployment    | $0.117 \pm 0.001$                                   | $0.019 \pm 0.002$        | $0.114 \pm 0.002$                     | $2.010 \pm 0.003$                     |
| Diabetes        | $0.032 \pm 0.002$                                   | $0.048 \pm 0.001$        | $0.022 \pm 0.002$                     | $2.132 \pm 0.001$                     |
| Hypertension    | $0.090 \pm 0.002$                                   | $0.183 \pm 0.003$        | $0.037 \pm 0.001$                     | $1.883 \pm 0.004$                     |
| ASSISTments     | $0.293 \pm 0.002$                                   | $0.260 \pm 0.001$        | $0.306 \pm 0.002$                     | $0.367 \pm 0.004$                     |

concept shift, improves the OOD test accuracy. We theoretically show that utilizing more informative variables helps maximizing predictive information by maximizing or minimizing certain terms in (1). **Proposition 4.3.** *If  $\phi_1(\cdot), \phi_2(\cdot)$  are invertible functions and if  $\mathbf{X}_I$  are informative variables such that  $Y \not\perp \mathbf{X}_I \mid \mathbf{X}, E$ , then we have the following inequalities:*

$$\begin{aligned}
 (i) & I(\phi_2(\{\mathbf{X} \cup \mathbf{X}_I\}); Y|E) > I(\phi_1(\mathbf{X}); Y|E) \quad (ii) I(\phi_2(\{\mathbf{X} \cup \mathbf{X}_I\}); E) > I(\phi_1(\mathbf{X}); E) \\
 (iii) & I(Y; E|\phi_2(\{\mathbf{X} \cup \mathbf{X}_I\})) < I(Y; E|\phi_1(\mathbf{X})) \quad (iv) I(\phi_2(\{\mathbf{X} \cup \mathbf{X}_I\}); E \mid Y) > I(\phi_1(\mathbf{X}); E \mid Y) \quad (5)
 \end{aligned}$$

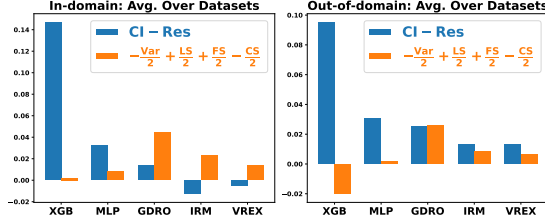
Proposition 4.3 shows that adding informative covariates increases conditional informativeness (5.i) and feature shift (5.ii), while reducing concept shift (5.iii). However, adding informative covariates also amplifies variation (5.iv), which may necessitate dedicated strategies to control the variation.

## 5 Experimental results

We conduct experiments on both real-world and synthetic datasets to analyze: (i) the presence of hidden confounding shift in real-world data, (ii) how the components of the decomposition in (1) affect OOD generalization under hidden confounding shifts, and (iii) the role of informative covariates in improving generalization. We consider eight real-world tabular benchmark datasets: *Food stamps*, *Readmission*, *Income*, *Public coverage*, *Unemployment*, *Diabetes*, *Hypertension*, and *ASSISTments*. These datasets and corresponding domain splits are adopted from *TableShift* benchmark [24]. We use ID test and OOD test accuracies to measure the performance of models. We perform experiments on two ERM-based methods: XGBoost (XGB) [15] and multi-layer perceptron (MLP), two domain generalization methods: IRM [6], VREX [38], and one robust learning based method: Group DRO (GDRO) [69]. We present average results over 5 different random hyperparameter choices as described in Appendix B. The effect of different random seeds is statistically insignificant for these datasets [24]. Additional details of the experimental setup are provided in Appendix § B. Results with hyperparameter tuning are presented in Appendix C.

**Hidden confounding shift:** Recall that hidden confounding shifts induce observable shifts in the distributions  $\mathbb{P}(\mathbf{X})$ ,  $\mathbb{P}(\mathbf{X} \mid Y)$ ,  $\mathbb{P}(Y)$ , and  $\mathbb{P}(Y \mid \mathbf{X})$ . Results in Table 2 show that label shift, covariate shift, conditional covariate shift, and concept shift are all present in real-world datasets, indicating the presence of hidden confounding shifts. For the qualitative analysis, we query GPT-4o [1] to list potential hidden confounders for several benchmark datasets [24, 57]. These results are presented in Appendix § E. For instance, in the *food stamps* dataset, unmeasured factors such as *economic policies* specific to each state may influence both *household* income and *food stamp reciprocity*.

**Conditional informativeness vs. accuracy:** From Proposition 4.2, under hidden confounding shift, maximizing the difference *conditional informativeness* – *residual* is essential for maximizing predictive information. Results on the first six datasets (that have more than one training environments [57, 24]) listed in Table 2 are shown in Figure 4. Dataset-wise results are in Appendix C. These results show that the difference *conditional informativeness* – *residual* is positively correlated with ID and OOD accuracy, with XGB being the best performing model with highest value of *conditional informativeness* – *residual*. As shown in the dataset-wise results in Appendix C, we observe that in many settings, due to hidden confounding shifts, the bars denoted with orange are close to zero because the sum  $-\text{variation} / 2 + \text{label shift} / 2 + \text{feature shift} / 2 - \text{concept shift} / 2$  is zero under hidden confounding shift (Proposition 4.2). The relative ID / OOD performance of the methods shown in Figure 1 is consistent with previously reported results in the literature [57, 24].



| Method | Avg. ID Test Acc. | Avg. OOD Test Acc. |
|--------|-------------------|--------------------|
| XGB    | <b>85.22</b>      | <b>77.00</b>       |
| MLP    | 83.64             | 73.29              |
| GDRO   | 78.40             | 72.28              |
| IRM    | 71.17             | 64.89              |
| VREX   | 59.74             | 63.06              |

Figure 4: The difference *conditional informativeness* – *residual* in the plots is positively correlated with the average ID and OOD test accuracy over the eight datasets shown in the table on the right.

Table 3: For all methods, sign consistency value is high for concept shift (CS). However, the sign consistency metric for conditional informativeness (CI) is crucial for generalization according to Proposition 4.2, and XGB excels at this. **Res**: residual, **Var**: variation, **FS**: feature shift, **C**: causal, **AC**: arguably causal, **A**: all.

| Method | Sign Consistency Metric ( $\uparrow$ ) |      |      |             |      | ID-Test Accuracy ( $\uparrow$ ) |              |              | OOD-Test Accuracy ( $\uparrow$ ) |              |              |
|--------|----------------------------------------|------|------|-------------|------|---------------------------------|--------------|--------------|----------------------------------|--------------|--------------|
|        | CI                                     | Var  | FS   | CS          | Res  | C                               | AC           | A            | C                                | AC           | A            |
| XGB    | <b>0.89</b>                            | 0.61 | 0.36 | <b>0.89</b> | 0.17 | <b>81.24</b>                    | <b>84.84</b> | <b>85.22</b> | <b>69.92</b>                     | <b>76.72</b> | <b>77.00</b> |
| MLP    | 0.75                                   | 0.64 | 0.28 | <b>0.89</b> | 0.33 | 80.51                           | 81.93        | 83.64        | 69.70                            | 73.84        | 73.29        |
| GDRO   | 0.69                                   | 0.67 | 0.19 | <b>0.94</b> | 0.36 | 80.92                           | 83.76        | 78.40        | 69.61                            | 73.49        | 72.28        |
| IRM    | 0.78                                   | 0.61 | 0.25 | <b>0.92</b> | 0.25 | 67.91                           | 68.22        | 71.17        | 61.21                            | 62.96        | 64.89        |
| VREX   | 0.64                                   | 0.61 | 0.25 | <b>0.89</b> | 0.44 | 54.39                           | 62.32        | 59.74        | 56.62                            | 63.53        | 63.06        |

**Informative covariates vs. accuracy:** We now study how inclusion of informative covariates helps in achieving better OOD generalization. To this end, we adopt the covariate partitioning from [57], where all available covariates are grouped into three nested covariate subsets: *causal covariates*  $\subseteq$  *arguably causal covariates*  $\subseteq$  *all covariates* (see Appendix § B). We evaluate how the terms in the decomposition in (1) affect ID and OOD performance when going from one covariate subset to another covariate subset. To this end, we use what we call the *sign consistency metric*, which works as follows. When we move from one covariate set to a larger covariate set, for any term in (1) with a positive coefficient, such as conditional informativeness, we count how often its value rises when accuracy improves. For negatively weighted terms, such as concept shift, we count how often its value decreases when accuracy improves. The metric is the fraction of observations where a term’s change aligns with its *beneficial* direction, thus capturing how reliably it contributes to better generalization. See Appendix § B for the formal definition. Table 3 shows that both conditional informativeness and concept shift exhibit high sign consistency with accuracy gains. While reducing concept shift is crucial, XGB further enhances conditional informativeness compared to other methods, yielding additional performance improvements. Dataset-specific results are presented in Appendix § C.

Note that in real-world datasets, it is often challenging to verify whether *all available covariates* include *every relevant or sufficiently informative* variable. To further test our hypothesis that informative covariates enhance generalization, we conduct experiments on synthetic data with a known causal structure:  $U \rightarrow X, U \rightarrow Y, X \rightarrow Y$ , and  $U \rightarrow \mathbf{X}_I$ , where  $\mathbf{X}_I$  denotes informative covariates caused by the hidden confounder  $U$ . We observe that including  $\mathbf{X}_I$  leads to improvements in conditional informativeness and feature shift, while reducing concept shift, as shown in Figure 5. Notably, the variation component remains consistently zero due to the simplicity of the problem. The synthetic data generation process and additional results are provided in Appendix § D.

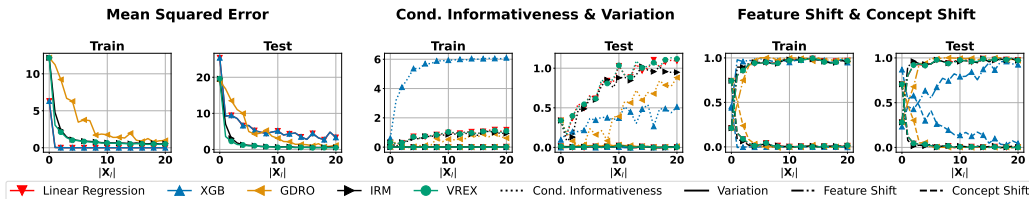


Figure 5: Adding more proxy variables  $\mathbf{X}_I$  of  $U$  that are informative to  $Y$  helps in reducing MSE, increasing conditional informativeness and feature shift while reducing concept shift.

## 6 Discussion, limitations, and future work

In summary, we decompose the predictive information between model outputs and true outcomes to reveal the factors limiting OOD generalization (Equation (1)). We argue that hidden confounding shifts—reflected in covariate and outcome distribution changes across environments—are unavoidable, and no finite set of observed covariates or proxies can fully resolve the induced ambiguity. Since confounding entangles covariate, label, concept, and feature shifts (inequalities in (2), (3)), treating these in isolation is ineffective. We show that learning environment-specific input–output mappings is more effective than enforcing a single invariant predictor (Equation (4)). By highlighting the inevitability of hidden confounding and the need to address it directly, our work lays a foundation for more robust OOD methods. Key challenges ahead include: (i) developing algorithms effective under limited confounder overlap, (ii) quantifying the cost–accuracy trade-off of acquiring informative covariates, (iii) handling entangled shifts without relying on untestable proxy assumptions, and (iv) modeling ambiguity from unobserved confounders to inspire new OOD-robust paradigms.

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## 678 Appendix

### 679 A Proofs of propositions

680 **Proposition 4.1.** For a covariate vector  $\mathbf{X}$ , label  $Y$ , with causal structure  $\mathbf{X} \leftrightarrow Y$ , i.e., some  
 681 covariates cause  $Y$  and some covariates are caused by  $Y$ , environment variable  $E$ , a feature  
 682 extractor  $\phi$ , and prediction  $\hat{Y}$ , the predictive information  $I(Y; \hat{Y})$  is decomposed as follows:

$$I(Y; \hat{Y}) = \overbrace{I(\phi(\mathbf{X}); Y|E)}^{\text{Cond. informativ.}} - \overbrace{\frac{I(\phi(\mathbf{X}); E|Y)}{2}}^{\text{Variation}} + \overbrace{\frac{I(Y; E)}{2}}^{\text{Label shift}} + \overbrace{\frac{I(\phi(\mathbf{X}); E)}{2}}^{\text{Feature shift}} - \overbrace{\frac{I(Y; E|\phi(\mathbf{X}))}{2}}^{\text{Concept shift}} - \overbrace{I(\phi(\mathbf{X}); Y|\hat{Y})}^{\text{Residual}} \quad (1)$$

683 where  $I(\phi(\mathbf{X}); Y|\hat{Y})$  is the residual information in  $\phi(\mathbf{X})$  for inferring  $Y$  that is not captured by the  
 684 prediction  $\hat{Y}$ . The decomposition above also holds when  $\phi(\mathbf{X})$  is replaced with  $\mathbf{X}$ .

685 *Proof.* From the causal graph shown in Figure 3 of the main paper, in the causal substructure:  
 686  $\hat{Y} \leftarrow \phi(\mathbf{X}) \leftarrow \mathbf{X} \leftrightarrow Y$ ,  $\hat{Y}$  provides no additional information in predicting  $Y$  given either  $\phi(\mathbf{X})$  or  
 687  $\mathbf{X}$ . This is because, the path  $\hat{Y} \leftarrow \phi(\mathbf{X}) \leftarrow \mathbf{X} \leftrightarrow Y$ , between  $\hat{Y}$  and  $Y$  is blocked by either  $\{\phi(\mathbf{X})\}$   
 688 or  $\{\mathbf{X}\}$ . This implies  $\hat{Y} \perp\!\!\!\perp Y|\phi(\mathbf{X})$  and  $\hat{Y} \perp\!\!\!\perp Y|\mathbf{X}$ . Now consider the expansion of  $I(Y; \hat{Y})$ .

$$\begin{aligned} I(Y; \hat{Y}) &= I(Y; \phi(\mathbf{X}), \hat{Y}) - I(\phi(\mathbf{X}); Y|\hat{Y}) \quad (\text{Introduce } \phi(\mathbf{X})) \\ &= I(\phi(\mathbf{X}); Y) + \underbrace{I(Y; \hat{Y}|\phi(\mathbf{X}))}_0 - I(\phi(\mathbf{X}); Y|\hat{Y}) \quad (\text{Expand and use } \hat{Y} \perp\!\!\!\perp Y | \phi(\mathbf{X})) \\ &= I(\phi(\mathbf{X}), E; Y) - I(Y; E|\phi(\mathbf{X})) - I(\phi(\mathbf{X}); Y|\hat{Y}) \quad (\text{Introduce } \mathbf{E}) \\ &= I(Y; E) + I(\phi(\mathbf{X}); Y|E) - I(Y; E|\phi(\mathbf{X})) - I(\phi(\mathbf{X}); Y|\hat{Y}) \quad (\text{Expand}) \end{aligned} \quad (6)$$

689 Similarly,

$$\begin{aligned} I(Y; \hat{Y}) &= I(Y; \phi(\mathbf{X}), \hat{Y}) - I(\phi(\mathbf{X}); Y|\hat{Y}) \quad (\text{Introduce } \phi(\mathbf{X})) \\ &= I(\phi(\mathbf{X}); Y) + \underbrace{I(Y; \hat{Y}|\phi(\mathbf{X}))}_0 - I(\phi(\mathbf{X}); Y|\hat{Y}) \quad (\text{Expand and use } \hat{Y} \perp\!\!\!\perp Y | \phi(\mathbf{X})) \\ &= I(Y, E; \phi(\mathbf{X})) - I(\phi(\mathbf{X}); E|Y) - I(\phi(\mathbf{X}); Y|\hat{Y}) \quad (\text{Introduce } \mathbf{E}) \\ &= I(\phi(\mathbf{X}); E) + I(\phi(\mathbf{X}); Y|E) - I(\phi(\mathbf{X}); E|Y) - I(\phi(\mathbf{X}); Y|\hat{Y}) \quad (\text{Expand}) \end{aligned} \quad (7)$$

690 Summing Equations (6) and (7) and grouping the terms gives the desired expression:

$$\begin{aligned} I(Y; \hat{Y}) &= I(\phi(\mathbf{X}); Y|E) - \frac{I(\phi(\mathbf{X}); E|Y)}{2} + \frac{I(Y; E)}{2} \\ &\quad + \frac{I(\phi(\mathbf{X}); E)}{2} - \frac{I(Y; E|\phi(\mathbf{X}))}{2} - I(\phi(\mathbf{X}); Y|\hat{Y}). \end{aligned}$$

691 To get the similar decomposition with  $\mathbf{X}$  instead of  $\phi(\mathbf{X})$ , use  $\hat{Y} \perp\!\!\!\perp Y | \mathbf{X}$  and introduce  $\mathbf{X}$  into the  
 692 expansion instead of  $\phi(\mathbf{X})$ .  $\square$

693 **Proposition 4.2.** For a covariate vector  $\mathbf{X}$ , label  $Y$ , an environment variable  $E$ , a feature extractor  
 694  $\phi$ , the prediction  $\hat{Y}$ , and an unobserved confounding variable  $U$  that cause both  $\mathbf{X}$  and  $Y$ , and if  
 695 either (i)  $\mathbf{X} \rightarrow Y$  or (ii)  $Y \rightarrow \mathbf{X}$ , the predictive information  $I(Y; \hat{Y})$  can be decomposed as follows:

$$I(Y; \hat{Y}) = \overbrace{I(\phi(\mathbf{X}); Y|E)}^{\text{Cond. informativeness}} - \overbrace{I(\phi(\mathbf{X}); Y|\hat{Y})}^{\text{Residual}} \quad (4)$$

696 *Proof.* Following the causal graph shown in Figure 3 of the main paper, we prove the proposition in  
 697 two separate cases:  $\mathbf{X} \rightarrow Y$  and  $Y \rightarrow \mathbf{X}$ .

698 **Case 1:**  $\mathbf{X} \rightarrow Y$ . In this case, we have the following inequality because conditioning on  $Y$  opens the  
 699 path  $\phi(\mathbf{X}) \leftarrow \mathbf{X} \rightarrow Y \leftarrow U \leftarrow E$  between  $\phi(\mathbf{X})$  and  $E$  at the collider node  $Y$ :

$$I(\phi(\mathbf{X}); E|Y) \geq I(\phi(\mathbf{X}); E). \quad (8)$$

700 Additionally, considering the causal substructure:  $Y \leftarrow \mathbf{X} \leftarrow U \leftrightarrow E$ , we have the following  
 701 inequality because conditioning on  $\phi(\mathbf{X})$  partially blocks the information flow from  $Y$  to  $E$  at the  
 702 node  $\mathbf{X}$  as long as  $\phi(\mathbf{X})$  encodes some information about  $\mathbf{X}$ :

$$I(Y; E) \geq I(Y; E|\phi(\mathbf{X})). \quad (9)$$

703 Now consider the following expansion from Equation (7) in the proof of Proposition 4.1.

$$\begin{aligned} I(Y; \hat{Y}) &= I(\phi(\mathbf{X}); E) + I(\phi(\mathbf{X}); Y|E) - I(\phi(\mathbf{X}); E|Y) - I(\phi(\mathbf{X}); Y|\hat{Y}) \quad (\text{Equation (7)}) \\ &\leq I(\phi(\mathbf{X}); Y|E) + I(\phi(\mathbf{X}); E|Y) - I(\phi(\mathbf{X}); E|Y) - I(\phi(\mathbf{X}); Y|\hat{Y}) \quad (\text{Using (8)}) \\ &= I(\phi(\mathbf{X}); Y|E) - I(\phi(\mathbf{X}); Y|\hat{Y}) \end{aligned}$$

704 Similarly, consider the following expansion from Equation (6) from the proof of Proposition 4.1:

$$\begin{aligned} I(Y; \hat{Y}) &= I(Y; E) + I(\phi(\mathbf{X}); Y|E) - I(Y; E|\phi(\mathbf{X})) - I(\phi(\mathbf{X}); Y|\hat{Y}) \quad (\text{Equation (6)}) \\ &\geq I(Y; E) + I(\phi(\mathbf{X}); Y|E) - I(Y; E) - I(\phi(\mathbf{X}); Y|\hat{Y}) \quad (\text{Using (9)}) \\ &= I(\phi(\mathbf{X}); Y|E) - I(\phi(\mathbf{X}); Y|\hat{Y}) \end{aligned}$$

705 That is, we have  $I(\phi(\mathbf{X}); Y|E) - I(\phi(\mathbf{X}); Y|\hat{Y}) \leq I(Y; \hat{Y}) \leq I(\phi(\mathbf{X}); Y|E) - I(\phi(\mathbf{X}); Y|\hat{Y})$ .

706 **Case 2:**  $Y \rightarrow \mathbf{X}$ . In this scenario, we have the following inequality because conditioning on  $\phi(\mathbf{X})$   
 707 opens the path  $E \rightarrow U \rightarrow \mathbf{X} \leftarrow Y$  between  $Y$  and  $E$  at the collider node  $\mathbf{X}$  because  $\phi(\mathbf{X})$  is a child  
 708 of  $\mathbf{X}$  [58].

$$I(Y; E|\phi(\mathbf{X})) \geq I(Y; E) \quad (10)$$

709 Additionally, we have the following inequality because conditioning on  $Y$  blocks the path  $\phi(\mathbf{X}) \leftarrow$   
 710  $\mathbf{X} \leftarrow Y \leftarrow U \leftrightarrow E$  at  $Y$  and hence less information flow between  $\phi(\mathbf{X})$  and  $E$ .

$$I(\phi(\mathbf{X}); E) \geq I(\phi(\mathbf{X}); E|Y) \quad (11)$$

711 Now consider the following expansion from Equation (6) from the proof of Proposition 4.1.

$$\begin{aligned} I(Y; \hat{Y}) &= I(Y; E) + I(\phi(\mathbf{X}); Y|E) - I(Y; E|\phi(\mathbf{X})) - I(\phi(\mathbf{X}); Y|\hat{Y}) \quad (\text{Equation (6)}) \\ &\leq I(\phi(\mathbf{X}); Y|E) + I(Y; E|\phi(\mathbf{X})) - I(Y; E|\phi(\mathbf{X})) - I(\phi(\mathbf{X}); Y|\hat{Y}) \quad (\text{Using (10)}) \\ &= I(\phi(\mathbf{X}); Y|E) - I(\phi(\mathbf{X}); Y|\hat{Y}) \end{aligned}$$

712 Similarly, consider the following expansion from Equation (7) from the proof of Proposition 4.1.

$$\begin{aligned} I(Y; \hat{Y}) &= I(\phi(\mathbf{X}); E) + I(\phi(\mathbf{X}); Y|E) - I(\phi(\mathbf{X}); E|Y) - I(\phi(\mathbf{X}); Y|\hat{Y}) \quad (\text{Equation (7)}) \\ &\geq I(\phi(\mathbf{X}); Y|E) + I(\phi(\mathbf{X}); E) - I(\phi(\mathbf{X}); E) - I(\phi(\mathbf{X}); Y|\hat{Y}) \quad (\text{Using (11)}) \\ &= I(\phi(\mathbf{X}); Y|E) - I(\phi(\mathbf{X}); Y|\hat{Y}) \end{aligned}$$

713 That is, we have  $I(\phi(\mathbf{X}); Y|E) - I(\phi(\mathbf{X}); Y|\hat{Y}) \leq I(Y; \hat{Y}) \leq I(\phi(\mathbf{X}); Y|E) - I(\phi(\mathbf{X}); Y|\hat{Y})$ .

714 Since the upper bound and lower bounds are the same in both cases, we have the desired result  
 715  $I(Y; \hat{Y}) = I(\phi(\mathbf{X}); Y|E) - I(\phi(\mathbf{X}); Y|\hat{Y})$ .  $\square$

716 **Proposition 4.3.** If  $\phi_1(\cdot), \phi_2(\cdot)$  are invertible functions and if  $\mathbf{X}_I$  are informative variables such that  
 717  $Y \not\perp \mathbf{X}_I \mid \mathbf{X}, E$ , then we have the following inequalities:

$$\begin{aligned} (i) & I(\phi_2(\{\mathbf{X} \cup \mathbf{X}_I\}); Y|E) > I(\phi_1(\mathbf{X}); Y|E) \quad (ii) I(\phi_2(\{\mathbf{X} \cup \mathbf{X}_I\}); E) > I(\phi_1(\mathbf{X}); E) \\ (iii) & I(Y; E|\phi_2(\{\mathbf{X} \cup \mathbf{X}_I\})) < I(Y; E|\phi_1(\mathbf{X})) \quad (iv) I(\phi_2(\{\mathbf{X} \cup \mathbf{X}_I\}); E \mid Y) > I(\phi_1(\mathbf{X}); E \mid Y) \end{aligned} \quad (5)$$

718 *Proof.* (i) By the chain rule of mutual information, we have:  $I(\mathbf{X}, \mathbf{X}_I; Y|E) = I(\mathbf{X}; Y|E) +$   
719  $I(\mathbf{X}_I; Y|E, \mathbf{X})$ . We have  $I(\mathbf{X}_I; Y|E, \mathbf{X}) > 0$  because  $Y \not\perp\!\!\!\perp \mathbf{X}_I \mid \mathbf{X}, E$ . Hence, we have  
720  $I(\mathbf{X}, \mathbf{X}_I; Y|E) > I(\mathbf{X}; Y|E)$ . Since  $\phi_1(\cdot), \phi_2(\cdot)$  are invertible functions and mutual information is  
721 invariant to invertible transformations, we have  $I(\phi_2(\{\mathbf{X} \cup \mathbf{X}_I\}); Y|E) > I(\phi_1(\mathbf{X}); Y|E)$ .

722 (ii) We consider the causal graph shown in Figure 3. From the chain rule of mutual information,  
723 we have:  $I(\mathbf{X}, \mathbf{X}_I; E) = I(\mathbf{X}; E) + I(\mathbf{X}_I; E|\mathbf{X})$ . Since  $E \leftrightarrow U \rightarrow Y$ , any covariate that is  
724 informative to  $Y$  is also informative to  $U$ . Since  $E \leftrightarrow U$ , we have  $I(\mathbf{X}_I; E|\mathbf{X}) > 0$ , and hence  
725  $I(\mathbf{X}, \mathbf{X}_I; E) > I(\mathbf{X}; E)$ . Since  $\phi_1(\cdot), \phi_2(\cdot)$  are invertible functions and mutual information is  
726 invariant to invertible transformations, we have  $I(\phi_2(\{\mathbf{X} \cup \mathbf{X}_I\}); E) > I(\phi_1(\mathbf{X}); E)$ .

727 (iii) Consider  $I(Y; E, \mathbf{X}_I | \mathbf{X})$ . It can be expressed in two ways:

$$I(Y; E, \mathbf{X}_I | \mathbf{X}) = I(Y; E | \mathbf{X}) + I(Y; \mathbf{X}_I | E, \mathbf{X}) = I(Y; \mathbf{X}_I | \mathbf{X}) + I(Y; E | \mathbf{X}_I, \mathbf{X})$$

728 Equating and rearranging the terms gives the following.

$$I(Y; E | \mathbf{X}) - I(Y; E | \mathbf{X}_I, \mathbf{X}) = I(Y; \mathbf{X}_I | \mathbf{X}) - I(Y; \mathbf{X}_I | E, \mathbf{X})$$

729 Since  $Y \not\perp\!\!\!\perp \mathbf{X}_I \mid E, \mathbf{X}$ , we have  $I(Y; \mathbf{X}_I | \mathbf{X}) > I(Y; \mathbf{X}_I | E, \mathbf{X})$  because conditioning usually  
730 reduces mutual information unless the additional conditioning variable opens any collider paths. Here  
731  $E$  is neither a collider nor a descendant of any collider. Thus,  $I(Y; E | \mathbf{X}) > I(Y; E | \mathbf{X}_I, \mathbf{X})$ . Since  
732  $\phi_1(\cdot), \phi_2(\cdot)$  are invertible functions and mutual information is invariant to invertible transformations,  
733 we have  $I(Y; E | \phi_1(\mathbf{X})) > I(Y; E | \phi_2(\{\mathbf{X}_I \cup \mathbf{X}\}))$ .

734 (iv) By the chain rule of mutual information, we have:  $I(\mathbf{X}, \mathbf{X}_I; E | Y) = I(\mathbf{X}; E | Y) +$   
735  $I(\mathbf{X}_I; E|\mathbf{X}, Y)$ . Since  $E \leftrightarrow U \rightarrow Y$ , any covariate that is informative to  $Y$  is also informative to  
736  $U$ . Since  $E \leftrightarrow U$ , we have  $I(\mathbf{X}_I; E|\mathbf{X}, Y) > 0$ , and hence  $I(\mathbf{X}, \mathbf{X}_I; E | Y) > I(\mathbf{X}; E | Y)$ . Since  
737  $\phi_1(\cdot), \phi_2(\cdot)$  are invertible functions and mutual information is invariant to invertible transformations,  
738 we have  $I(\phi_2(\{\mathbf{X} \cup \mathbf{X}_I\}); E | Y) > I(\phi_1(\mathbf{X}); E | Y)$ .  $\square$

## 739 B Experimental setup

740 In this section, we detail our experimental setup. The impact of random seeds on the results is  
741 statistically insignificant in many settings [24]. For the results in the main paper, we report the mean  
742 and standard deviation across five random hyperparameter values (sampled from their respective  
743 domains in Table B1) for each dataset-method combination. Since we evaluate models on ID test and  
744 OOD test data, to evaluate mutual information terms, we use the nonparametric entropy estimation  
745 toolbox [37, 77, 78]. Following [24], we evaluate mutual information using both in-distribution (ID)  
746 test data (from training domains) and out-of-distribution (OOD) test data (from test domains).

747 For consistency, we select 20,000 random samples  
748 from each ID and OOD test data, though  
749 some datasets contain fewer than 20,000 sam-  
750 ples. Throughout our experiments, the sam-  
751 ple size has a relatively insignificant effect on  
752 the evaluated mutual information terms. For  
753 instance, as shown in Figure B1, the terms in  
754 Equation (1) computed for the representations  
755 of the XGB model on the *income* dataset stayed  
756 relatively consistent as we vary sample size. Fea-  
757 tures  $\phi(\mathbf{X})$  are extracted from the layer preced-  
758 ing the classification head; for XGBoost, we  
759 combine output margins and SHAP values from the model. All experiments were conducted on a  
760 single NVIDIA RTX A6000 GPU.

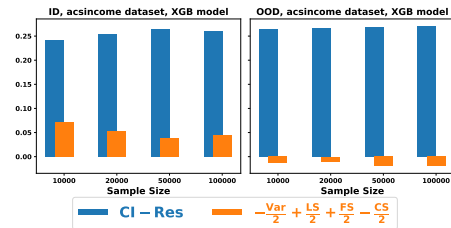


Figure B1: Evaluated terms of predictive information are relatively consistent across various sample sizes on the *income* dataset.

761 In our experiments studying the impact of informative covariates on predictive information, we follow  
762 the categorization introduced by Nastl and Hardt [57], partitioning covariates into: *causal*, *arguably*  
763 *causal*, and *all covariates*. Causal covariates have clearly established influence on the target, with well-  
764 supported directionality and unlikely reverse causation. However, strict reliance on causal variables  
765 risks omitting relevant parents due to knowledge gaps. Arguably causal covariates have uncertain  
766 causal relationships, meeting at least one of: (1) being known causal, (2) having plausible but poten-  
767 tially bidirectional influence, or (3) likely (but unconfirmed) causal effect. These groups approximate

Table B1: Hyperparameters and their possible values considered in this paper.

| Model    | Hyperparameter                    | Search values         |
|----------|-----------------------------------|-----------------------|
| XGBoost  | Learning rate                     | [0.1, 0.2, 0.3]       |
|          | Maximum tree depth                | [4, 5, 6]             |
|          | Minimum child weight              | [0.1, 1, 10]          |
|          | Gamma (min. loss reduction)       | [0.0001, 0.001, 0.01] |
|          | Subsample ratio                   | [0.5, 0.6, 0.7]       |
|          | Column subsample per tree         | [0.5, 0.6, 0.7]       |
|          | L1 regularization ( $\alpha$ )    | [0.0001, 0.001, 0.01] |
|          | L2 regularization ( $\lambda$ )   | [0.0001, 0.001, 0.01] |
| MLP      | Number of layers                  | [2, 3, 4]             |
|          | Hidden layer size                 | [256, 512, 1024]      |
|          | Dropout rate                      | [0.0, 0.1, 0.2]       |
|          | Learning rate                     | [0.01, 0.02, 0.05]    |
|          | Weight decay (L2)                 | [0.0001, 0.001, 0.01] |
|          | Batch size                        | [4096]                |
|          | Number of epochs                  | [1, 2, 3]             |
| GroupDRO | Number of layers                  | [2, 3, 4]             |
|          | Hidden layer size                 | [256, 512, 1024]      |
|          | Group-weights step size           | [0.01, 0.05, 0.1]     |
|          | Dropout rate                      | [0.0, 0.1, 0.2]       |
|          | Learning rate                     | [0.01, 0.02, 0.05]    |
|          | Weight decay (L2)                 | [0.0001, 0.001, 0.01] |
|          | Batch size                        | [4096]                |
| IRM      | Number of layers                  | [2, 3, 4]             |
|          | Hidden layer size                 | [256, 512, 1024]      |
|          | Dropout rate                      | [0.0, 0.1, 0.2]       |
|          | IRM penalty weight ( $\lambda$ )  | [0.01, 0.05, 0.1]     |
|          | IRM penalty anneal iterations     | [1, 2, 3]             |
|          | Learning rate                     | [0.01, 0.02, 0.05]    |
|          | Weight decay (L2)                 | [0.0001, 0.001, 0.01] |
|          | Batch size                        | [4096]                |
| VREX     | Number of layers                  | [2, 3, 4]             |
|          | Hidden layer size                 | [256, 512, 1024]      |
|          | VREX penalty anneal iterations    | [1, 2, 3]             |
|          | VREX penalty weight ( $\lambda$ ) | [0.1, 10, 100]        |
|          | Dropout rate                      | [0.0, 0.1, 0.2]       |
|          | Learning rate                     | [0.01, 0.02, 0.05]    |
|          | Weight decay (L2)                 | [0.0001, 0.001, 0.01] |
|          | Batch size                        | [4096]                |
| VREX     | Number of layers                  | [2, 3, 4]             |
|          | Hidden layer size                 | [256, 512, 1024]      |
|          | VREX penalty anneal iterations    | [1, 2, 3]             |
|          | VREX penalty weight ( $\lambda$ ) | [0.1, 10, 100]        |
|          | Dropout rate                      | [0.0, 0.1, 0.2]       |
|          | Learning rate                     | [0.01, 0.02, 0.05]    |
|          | Weight decay (L2)                 | [0.0001, 0.001, 0.01] |
|          | Batch size                        | [4096]                |
| VREX     | Number of layers                  | [2, 3, 4]             |
|          | Hidden layer size                 | [256, 512, 1024]      |
|          | VREX penalty anneal iterations    | [1, 2, 3]             |
|          | VREX penalty weight ( $\lambda$ ) | [0.1, 10, 100]        |
|          | Dropout rate                      | [0.0, 0.1, 0.2]       |
|          | Learning rate                     | [0.01, 0.02, 0.05]    |
|          | Weight decay (L2)                 | [0.0001, 0.001, 0.01] |
|          | Batch size                        | [4096]                |

768 true causal parents based on available knowledge, though relationships may be confounded. Some  
769 datasets additionally contain *anti-causal* covariates where the target likely affects them but not vice  
770 versa. The *all covariates* set includes all observed variables regardless of causal status. The sign con-  
771 sistency metric introduced in the main paper § 5 is defined as follows. For each  $(m, \sigma_m) \in M$  where  
772  $M = \{(\text{conditional informativeness}, +1), (\text{variation}, -1), (\text{label shift}, +1), (\text{feature shift}, +1),$   
773  $(\text{concept shift}, -1), \text{ and } (\text{residual}, -1)\}$ , for a specific dataset, we compute:

$$C_m = \frac{1}{2|P|} \sum_{(s,t) \in P} [\mathbb{I}(\sigma_m \cdot (\mu_t^{\text{tr}}(m) - \mu_s^{\text{tr}}(m)) > 0) + \mathbb{I}(\sigma_m \cdot (\mu_t^{\text{te}}(m) - \mu_s^{\text{te}}(m)) > 0)]$$

774 Where  $P$  is the set of covariate set pairs  $(s, t)$  with  $t \supset s$ . For example,  $s$  can be causal covariates  
775 and  $t$  can be arguably causal covariates. There is a factor of 2 in  $2P$  that accounts for both train and

test data.  $\mu_s^{tr/te}(m)$  is the mean of measure  $m$  for covariate set  $s$  in training/testing data.  $\mathbb{I}(\cdot)$  is the indicator function.  $|P| = 3$  for covariate settings  $\{causal, arguably\ causal, all\}$ . To get final sign consistency metric value for each  $m$ , we average across datasets:  $\bar{C}_m = \frac{1}{|D|} \sum_{d \in D} C_m^{(d)}$  where  $|D|$  is the number of datasets.

## C Additional results on real-world datasets

In this section, we present additional results on real-world datasets. Figures C2, C3 show the comparison of the terms: *conditional informativeness* – *residual* and *–variation* / 2 + *label shift* / 2 + *feature shift* / 2 – *concept shift* / 2 for each dataset (their aggregation is in the main paper Figure 4 Left). The key takeaway from these results is that, from Proposition 4.1, the sum of blue and orange terms is positively correlated with the overall model performance. At the same time, due to potential hidden confounding shift in real-world data, the contribution of the difference *conditional informativeness* – *residual* is higher towards the overall predictive information when compared to the contribution of *–variation* / 2 + *label shift* / 2 + *feature shift* / 2 – *concept shift* / 2. This serves as a motivation to maximize conditional informativeness for better OOD generalization under a hidden confounding shift.

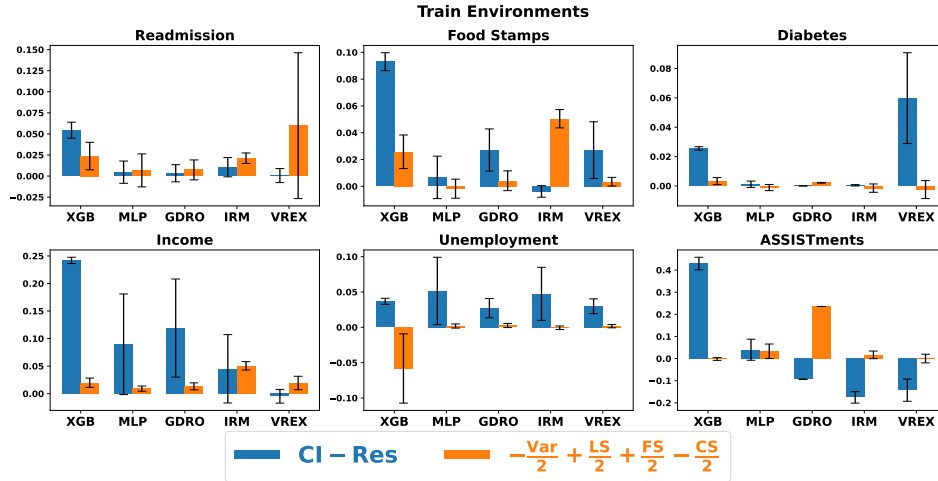


Figure C2: Dataset specific results on ID data.

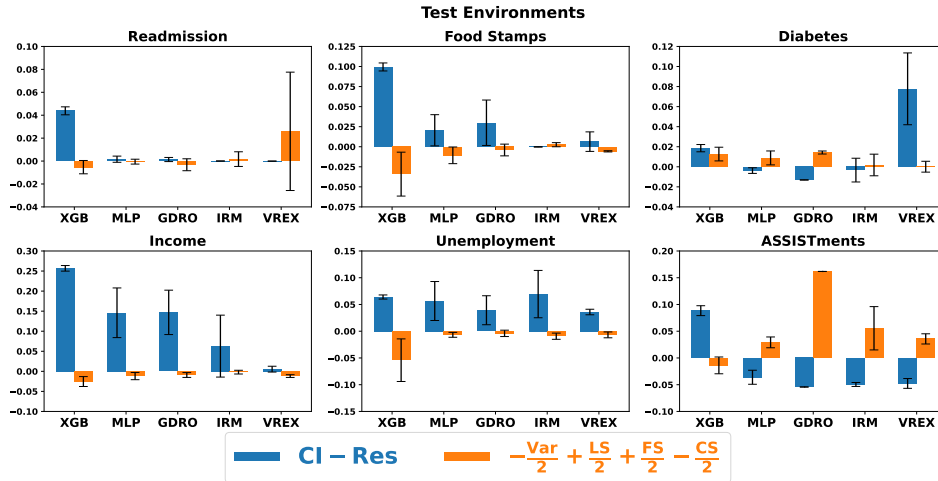


Figure C3: Dataset-specific results on OOD data.

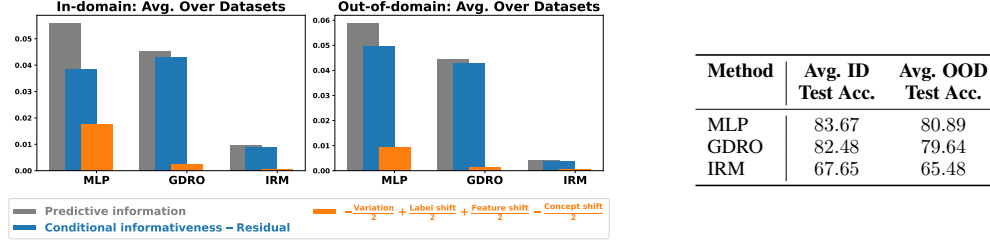


Figure C4: The difference *conditional informativeness* – *residual* in the plots is positively correlated with the average ID and OOD test accuracy over five datasets shown in the table on the right. The contribution of the difference *conditional informativeness* – *residual* is higher towards the overall predictive information when compared to the contribution of  $-\text{variation} / 2 + \text{label shift} / 2 + \text{feature shift} / 2 - \text{concept shift} / 2$ .

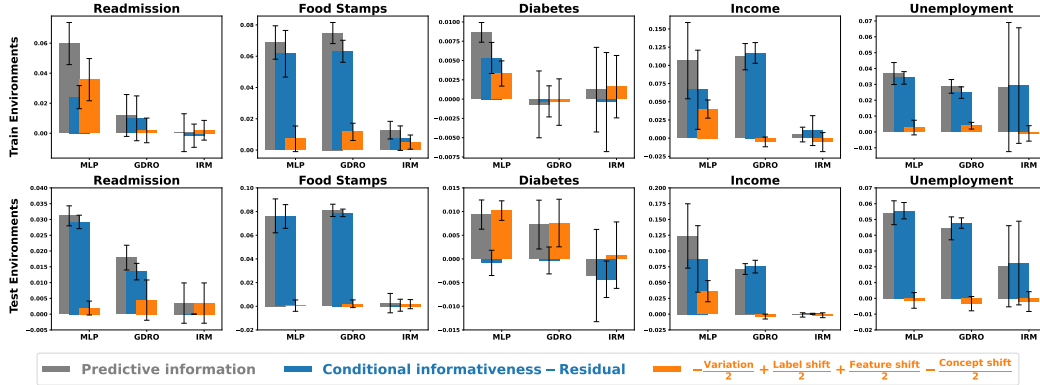


Figure C5: Dataset specific results. Comparison of the terms in predictive information.

Table C2: Comparison of ID and OOD test accuracies of models.

| Method | Readmission  |               | Food stamps  |               | Diabetes     |               |
|--------|--------------|---------------|--------------|---------------|--------------|---------------|
|        | ID test acc. | OOD test acc. | ID test acc. | OOD test acc. | ID test acc. | OOD test acc. |
| MLP    | 65.78 ± 0.14 | 61.75 ± 0.22  | 84.74 ± 0.03 | 82.00 ± 0.06  | 87.66 ± 0.02 | 83.22 ± 0.05  |
| GDRO   | 60.47 ± 1.08 | 57.60 ± 0.42  | 84.47 ± 0.04 | 81.41 ± 0.09  | 87.47 ± 0.10 | 82.90 ± 0.10  |
| IRM    | 50.28 ± 7.35 | 50.85 ± 1.74  | 80.91 ± 0.00 | 78.01 ± 0.00  | 42.54 ± 36.5 | 43.48 ± 31.9  |
|        | Income       |               | Unemployment |               |              |               |
|        | ID test acc. | OOD test acc. | ID test acc. | OOD test acc. |              |               |
| MLP    | 82.92 ± 0.05 | 81.46 ± 0.01  | 97.27 ± 0.02 | 96.04 ± 0.05  |              |               |
| GDRO   | 82.70 ± 0.05 | 80.16 ± 0.45  | 97.28 ± 0.01 | 96.10 ± 0.00  |              |               |
| IRM    | 67.91 ± 0.00 | 60.20 ± 0.00  | 96.61 ± 0.00 | 94.84 ± 0.00  |              |               |

Figures C6- C15 further show the comparison of various terms of the information decomposition in (1) for each method and dataset. From these results, we observe that the values plotted in the second column (*variation*) and third column (*feature shift*) are similar in magnitude, and based on Equation 1, their difference is close to zero, contributing very little to the overall predictive decomposition. Similarly, we observe that the values plotted in the fourth column (*label shift*) and fifth column (*concept shift*) are similar in magnitude, and based on Equation 1, their difference is close to zero, contributing very little to the overall predictive decomposition. In almost all settings, the difference *conditional informativeness* - *residual* is the majority contributor for the predictive information.

**Hyperparameter tuning:** We also perform hyperparameter tuning on model parameters. We run a grid search over hyperparameter values and select the best hyperparameter values based on ID validation accuracy (not OOD validation accuracy). We run 1500 experiments (5 datasets, 3 models, and 100 hyperparameter choices sampled from a grid of hyperparameter values). In this case, to get the mean and standard deviation results for accuracies and mutual information terms, we run

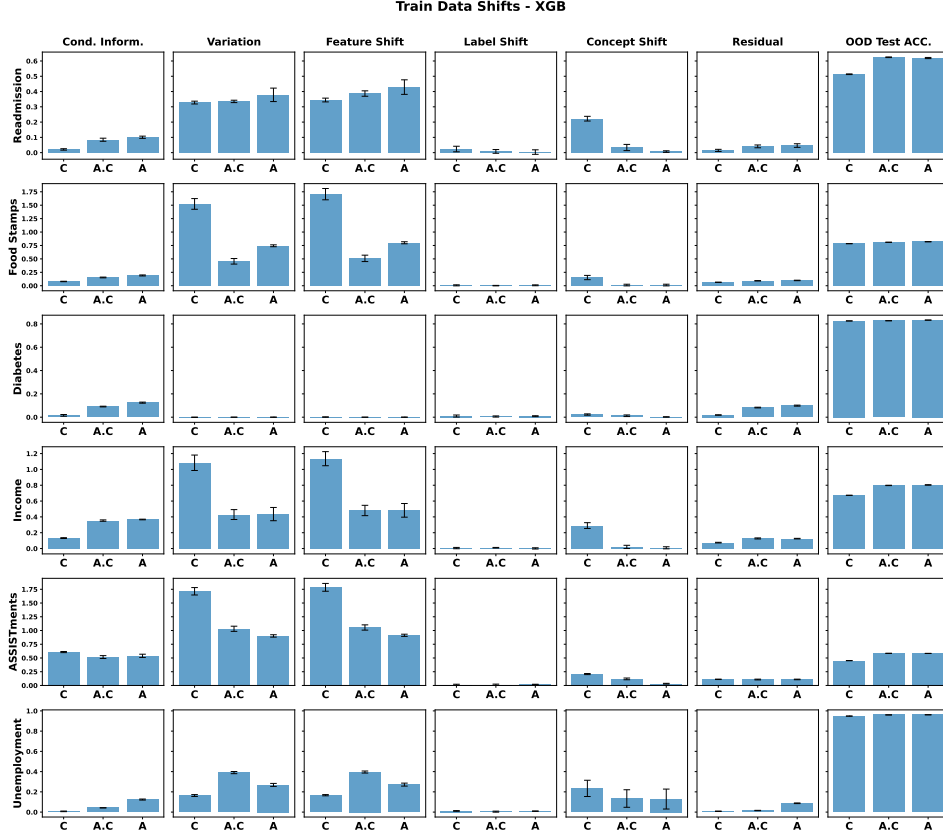


Figure C6: Decomposition of information metrics on train data for XGBoost and groups of features: (C) Causal, (A.C) Arguably causal, (A) All.

each experiment for 5 random seeds, even if the effect of random seeds is insignificant in many cases. Results are shown in Figure C4. From Proposition 4.1, the sum of blue and orange terms is positively correlated with the overall model performance. At the same time, due to potential hidden confounding shift in real-world data, the contribution of the difference *conditional informativeness* – *residual* is higher towards the overall predictive information when compared to the contribution of  $-\text{variation} / 2 + \text{label shift} / 2 + \text{feature shift} / 2 - \text{concept shift} / 2$ . Dataset-specific results shown in Figure C5 and Table C2 also show similar trends.

**Potential limitations with predictive information decomposition analysis:** While predictive information  $I(Y; \hat{Y})$  measures statistical dependence between predictions and labels, it does not guarantee accuracy. For instance, a binary classifier that systematically predicts the *wrong* label (e.g.,  $\hat{Y} = 1$  when  $Y = 0$ ) achieves maximal  $I(Y; \hat{Y}) = 1$  bit (for binary variables) due to perfect anti-correlation, yet yields 0% accuracy. This occurs because mutual information quantifies *reduction in uncertainty* rather than correctness. Other decomposition terms provide further insight: conditional informativeness  $I(\phi(X); Y|E)$  may mask harmful variation  $I(\phi(X); E|Y)$ , while feature shift  $I(\phi(X); E)$  can inflate  $I(Y; \hat{Y})$  through spurious correlations. Nevertheless, this decomposition remains valuable for analyzing reasonably performant models, as our experiments demonstrate - it reveals whether predictive information stems from generalizable patterns or environment-specific artifacts, enabling targeted improvements while maintaining interpretability.

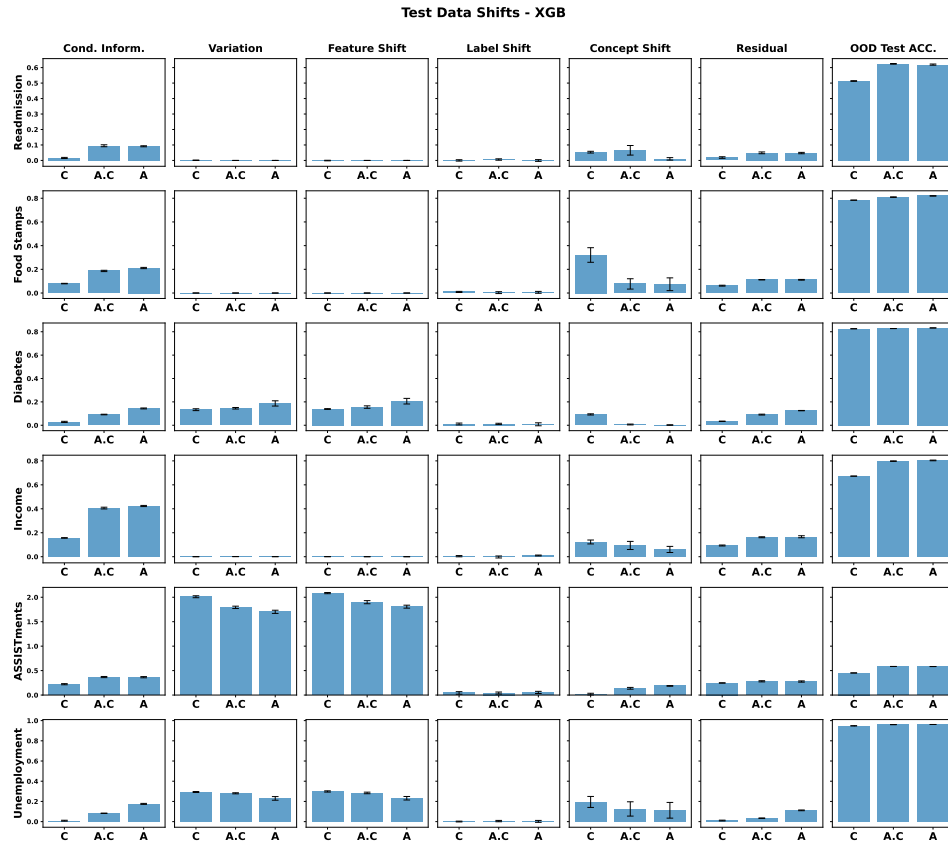


Figure C7: Decomposition of information metrics on test data for XGBoost and groups of features: (C) Causal, (A.C) Arguably causal, (A) All.

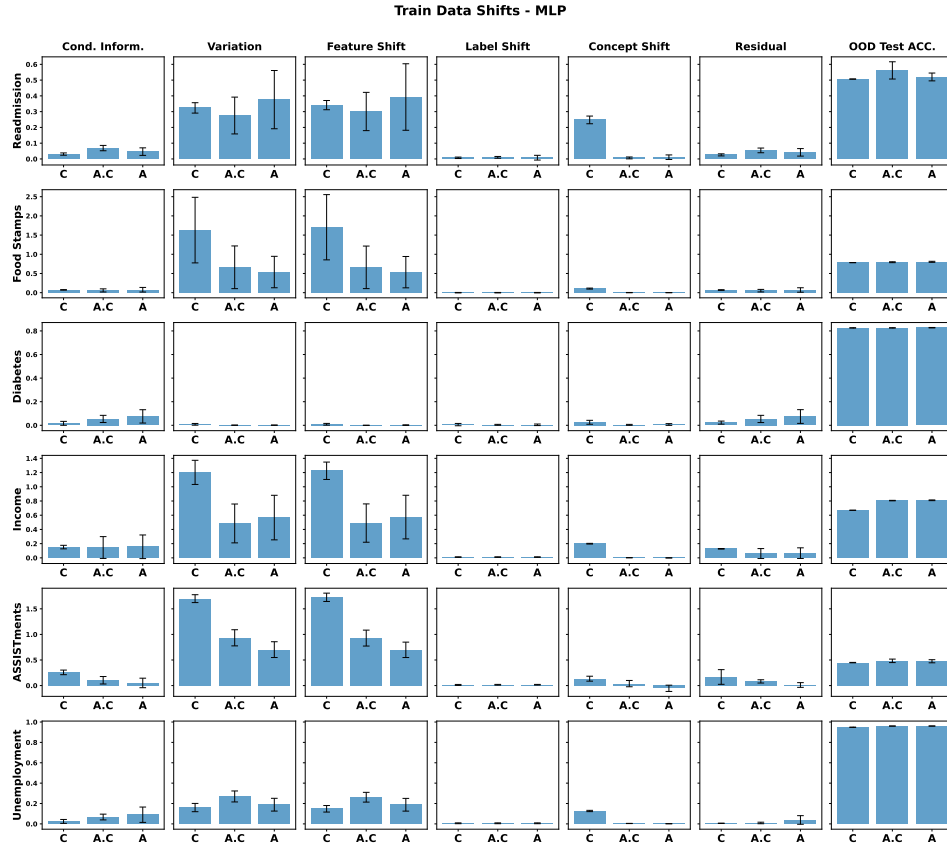


Figure C8: Decomposition of information metrics on train data for MLP and groups of features: (C) Causal, (A.C) Arguably causal, (A) All.

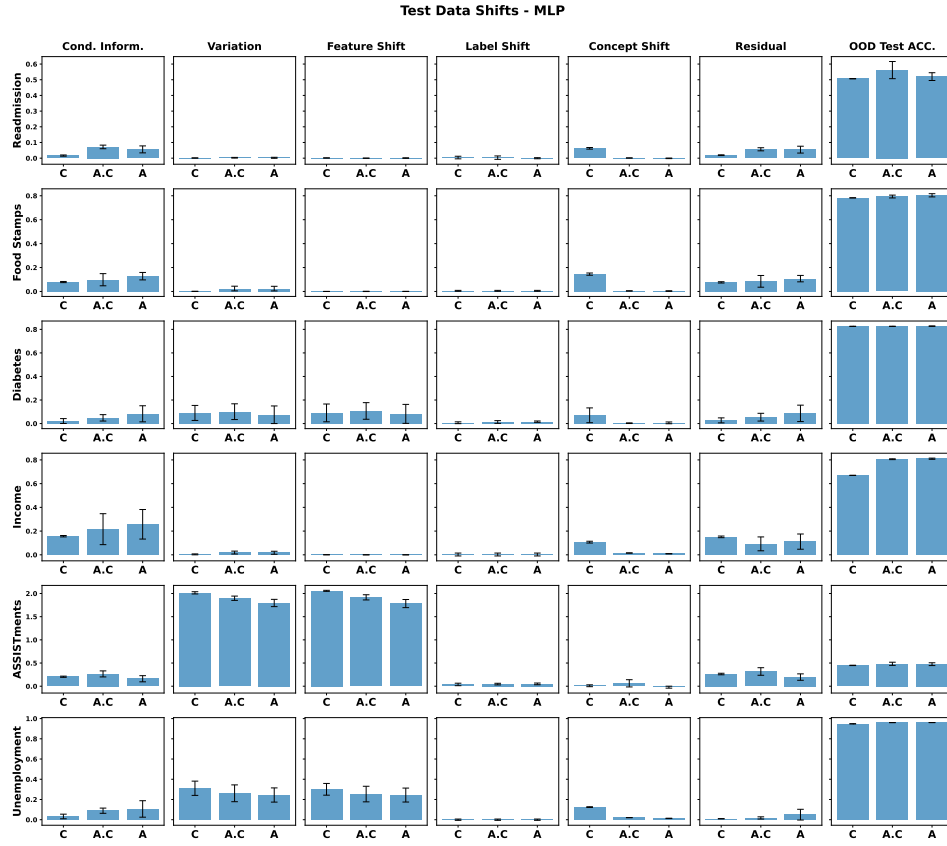


Figure C9: Decomposition of information metrics on test data for MLP and groups of features: (C) Causal, (A.C) Arguably causal, (A) All.

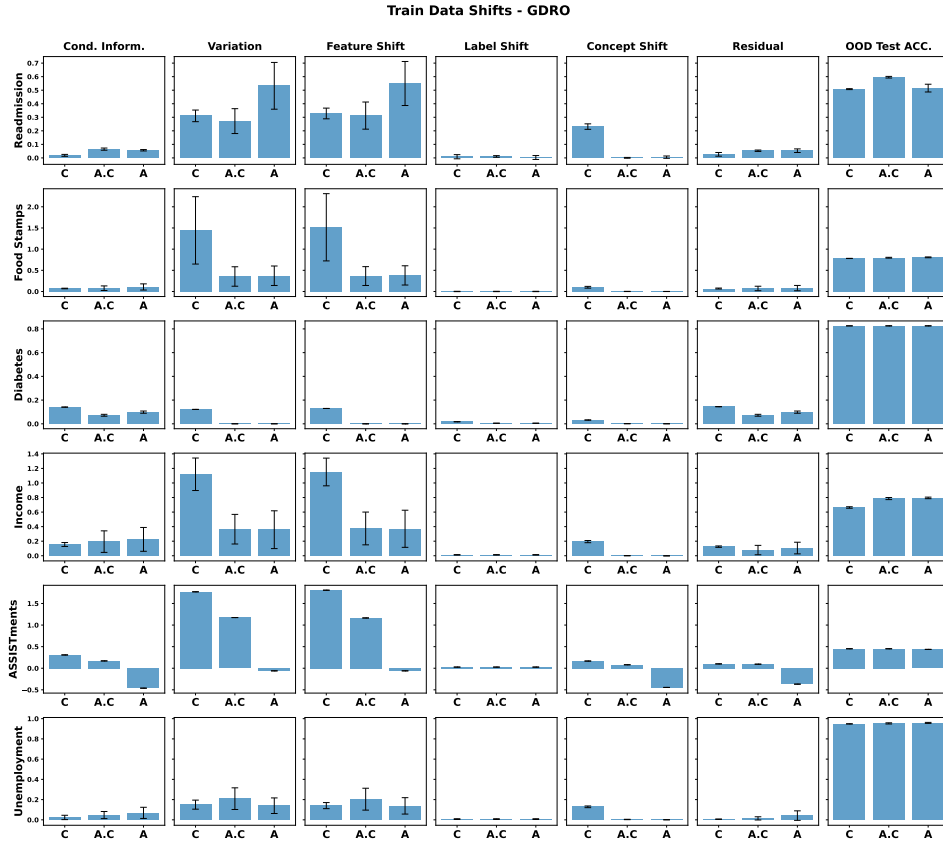


Figure C10: Decomposition of information metrics on train data for GDRO and groups of features: (C) Causal, (A.C) Arguably causal, (A) All.

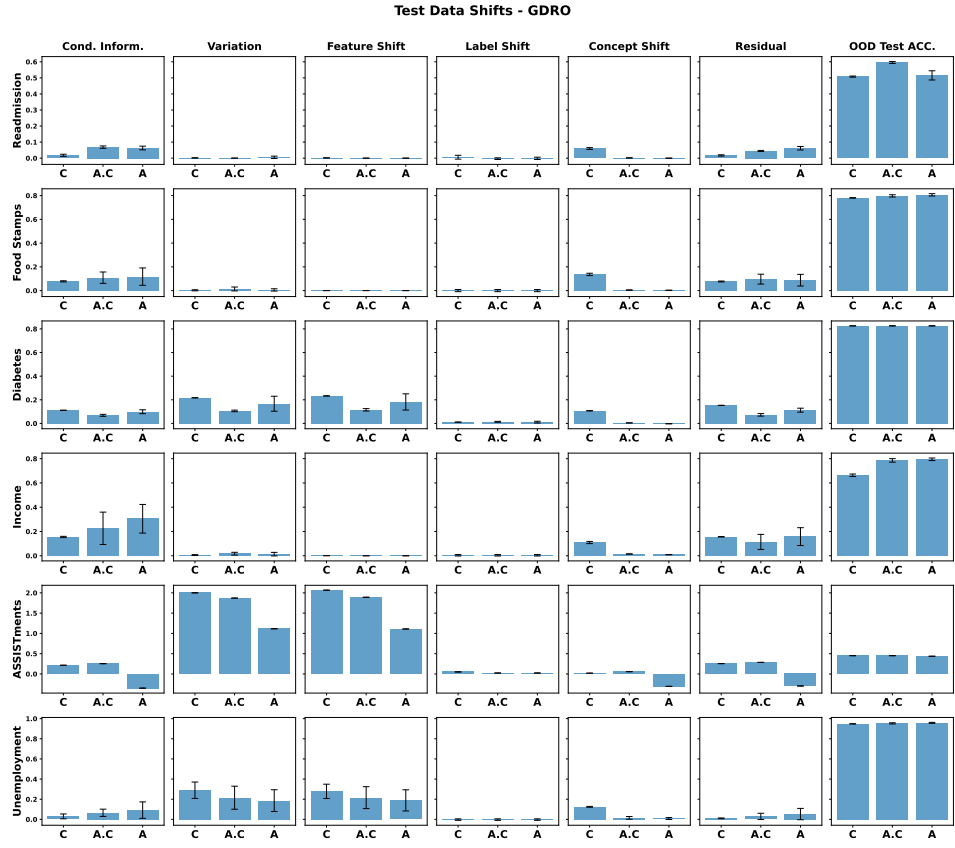


Figure C11: Decomposition of information metrics on test data for GDRO and groups of features: (C) Causal, (A.C) Arguably causal, (A) All.

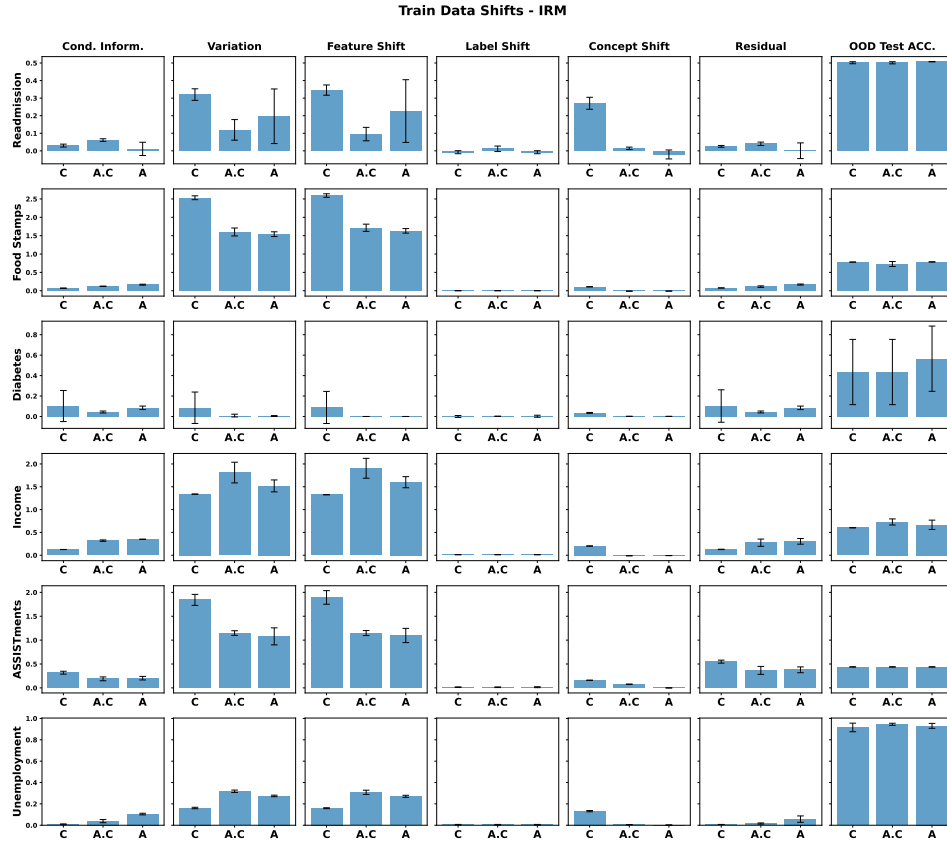


Figure C12: Decomposition of information metrics on train data for IRM and groups of features: (C) Causal, (A.C) Arguably causal, (A) All.

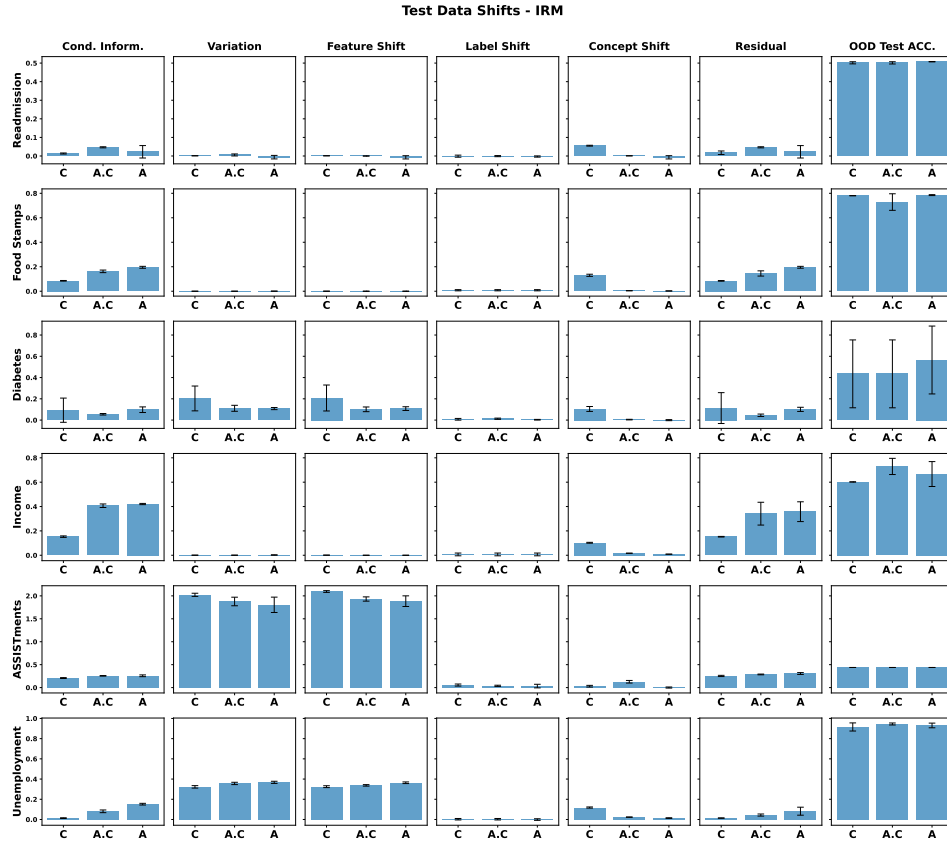


Figure C13: Decomposition of information metrics on test data for IRM and groups of features: (C) Causal, (A.C) Arguably causal, (A) All.

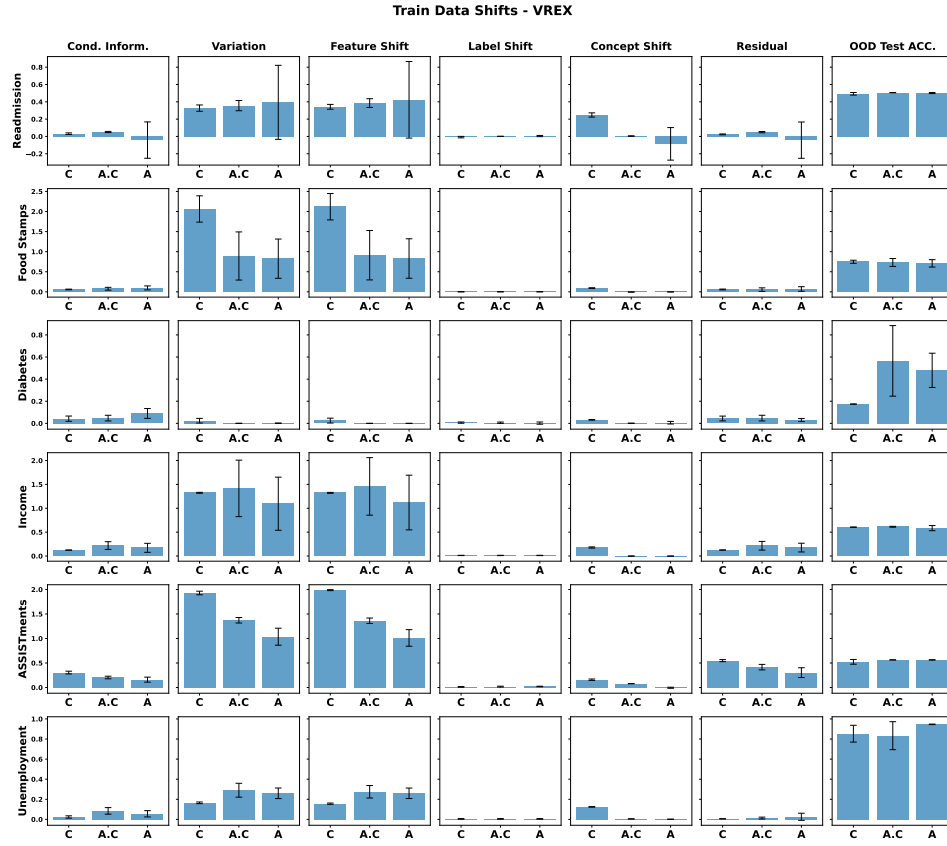


Figure C14: Decomposition of information metrics on train data for VREX and groups of features: (C) Causal, (A.C) Arguably causal, (A) All.

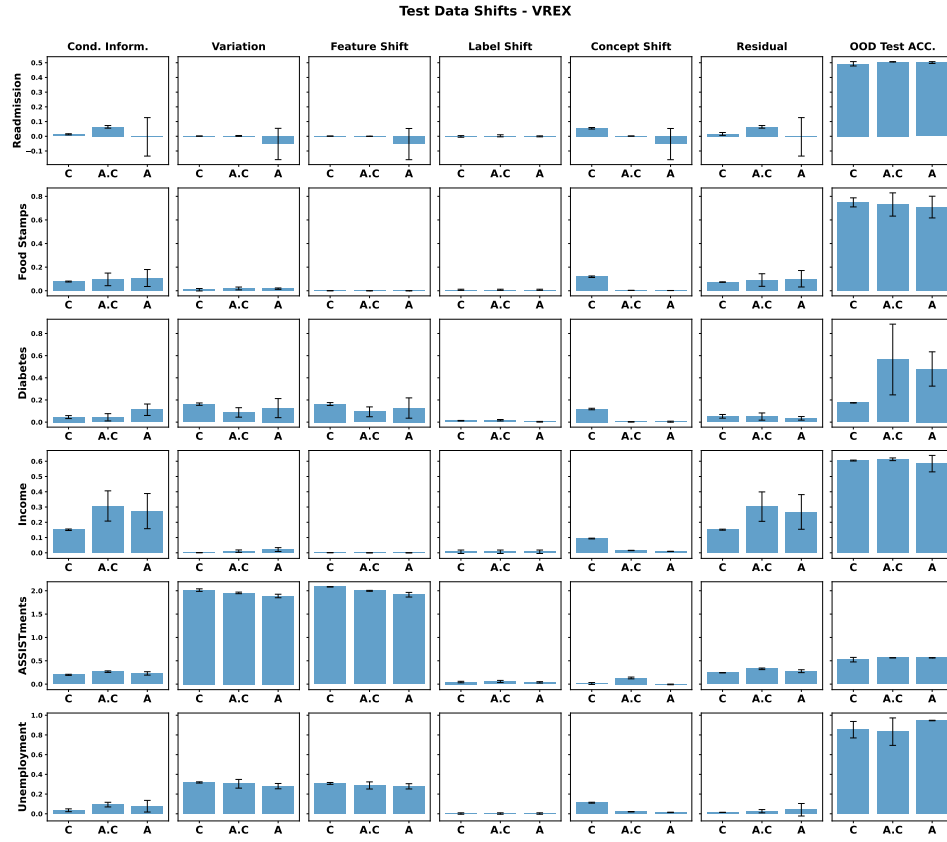


Figure C15: Decomposition of information metrics on test data for VREX and groups of features: (C) Causal, (A.C) Arguably causal, (A) All.

## 822 D Results on synthetic datasets

823 We conduct several experiments on synthetic datasets to investigate: (i) the impact of different  
 824 informativeness criteria, (ii) how the level of confounder overlap across environments affects model  
 825 performance, and (iii) the influence of the number of causal versus anti-causal variables on the  
 826 performance. The data-generating process follows the structural equations in (12), with the causal  
 827 structure  $U \rightarrow X, U \rightarrow Y, U \rightarrow \mathbf{X}_I$ , and  $X \rightarrow Y$ , where  $U$  is an unobserved confounder,  $X$  is an  
 828 observed covariate,  $Y$  is the target variable, and  $\mathbf{X}_I$  represents additional informative covariates.

$$\begin{aligned} U &\sim \mathcal{N}(\mu_u^e, \sigma_u) & X &\leftarrow f_X(U) + \mathcal{N}(\mu_x, \sigma_x) \\ X_i &\leftarrow f_i(U) + \mathcal{N}(\mu_i, \sigma_i); X_i \in \mathbf{X}_I & Y &\leftarrow f_Y(X, U) + \mathcal{N}(\mu_y, \sigma_y) \end{aligned} \quad (12)$$

829 We explore different functional forms for  $f_X$ ,  $f_i$ , and  $f_Y$ , with the corresponding results presented in  
 830 our analysis. Domain shifts are induced by systematically varying the environment-specific mean  
 831 parameter  $\mu_u^e$  of the confounder distribution. This allows us to examine how different degrees of  
 832 distributional shift affect model performance across environments.

833 **Informativeness vs. accuracy:** In this set of experiments, we begin with linear structural equations:  
 834  $f_X = 0.3U$ ,  $f_i = 0.1U$ , and  $f_Y = X - 2U$ . For these experiments  $\mu_u^e \in \{-2, 2\}$  for training  
 835 environments and  $\mu_u^e \in \{0, 4\}$  for test environments. For the results presented in Figure 5 of the  
 836 main paper, we consider one hidden confounding variable with  $|\mathbf{X}_I| = 20$  informative covariates.  
 837 As we increase the number of informative covariates from 0 to 20, we observe: (i) a reduction in  
 838 mean squared error, (ii) improved conditional informativeness, (iii) enhanced feature shift, while  
 839 (iv) decreased concept shift. We extend these findings by examining two additional informativeness  
 840 criteria. First, we analyze the case with one hidden confounder and a single informative covariate  
 841 ( $|\mathbf{X}_I| = 1$ ) across varying noise levels  $\sigma_i \in \{0, 0.1, \dots, 2.0\}$ . Figure D16 demonstrates that as noise  
 842 decreases from 2.0 to 0.0, we observe: (i) reduced mean squared error, (ii) improved conditional  
 843 informativeness, (iii) enhanced feature shift, alongside (iv) decreased concept shift. Consistent with  
 844 other synthetic experiments, the variation term remains zero throughout. Finally, we investigate a  
 845 setting with 20 hidden confounders and corresponding 20 informative covariates, using modified  
 846 structural equations  $f_X = 0.3U$ ,  $f_i = 0.2U$ , and  $f_Y = X - 2U$ . Figure D17 shows that as we  
 847 introduce informative covariates corresponding to hidden confounders, we again observe: (i) reduced  
 848 mean squared error, (ii) improved conditional informativeness, (iii) enhanced feature shift, while  
 849 (iv) decreased concept shift. These experiments demonstrate how different informativeness criteria  
 850 help in achieving OOD generalization.

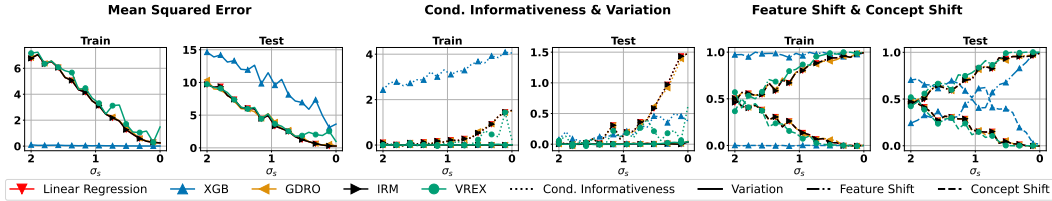


Figure D16: Adding proxy variable with low noise helps in reducing MSE, increasing conditional informativeness and feature shift while reducing concept shift.

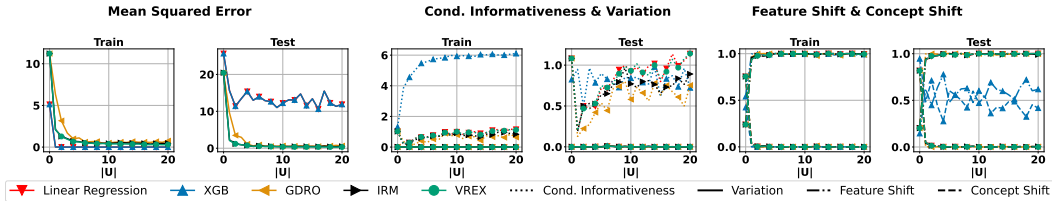


Figure D17: Adding more proxy variables  $\mathbf{X}_I$  of a set of hidden confounding variables  $\mathbf{U}$  that are informative to  $Y$  helps in reducing MSE, increasing conditional informativeness and feature shift while reducing concept shift.

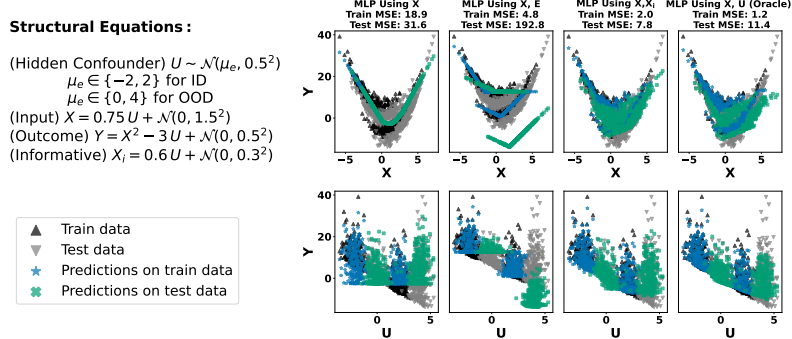


Figure D18: Performance of MLP on the low overlap non-linear setting, for sets of features: (i)  $X$  (ii)  $X$  and  $E$  (environment statistics) (iii)  $X$  and  $X_i$  (iv)  $X$  and  $U$  (Oracle).

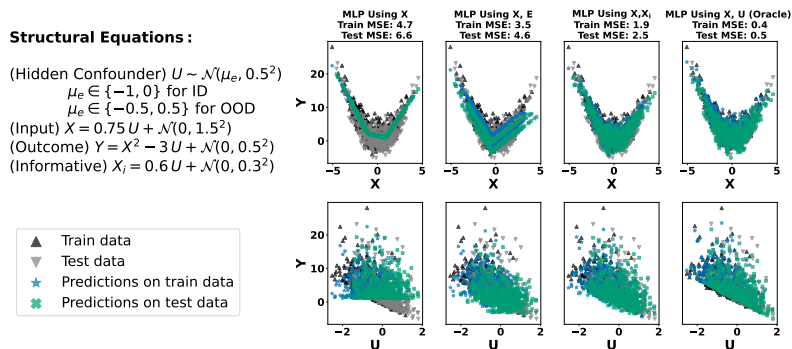


Figure D19: Performance of MLP on the high overlap non-linear setting, for sets of features: (i)  $X$  (ii)  $X$  and  $E$  (environment statistics) (iii)  $X$  and  $X_i$  (iv)  $X$  and  $U$  (Oracle).

**Overlapping confounder support vs. accuracy:** As established in the main paper, recent work by Prashant et al. [62] proposed an OOD generalization method for hidden confounding shift that assumes test confounder support remains within training support. While our results demonstrated successful learning of correct relationships in synthetic linear data when sufficient information about hidden confounders exists in observed covariates, we now extend this analysis to nonlinear data using MLP and XGB models under both low and high confounding support conditions (Figures D18–D21). In low confounding overlap settings (characterized by distant  $\mu_e$  between ID and OOD data), models relying solely on  $X$  learn an inadequate global function that fails to distinguish environments. Performance improves when incorporating environment-specific summary statistics of observed covariates, which helps capture environment-specific relationships, but the most significant gains occur only when leveraging additional informative covariates, underscoring their importance. For completeness, we include both oracle model results and  $U$ - $Y$  relationship scatter plots to elucidate the learned input-output mappings. These experiments confirm better performance under high confounding overlap as expected, but more importantly, reveal promising results in the low-overlap regime - a previously unstudied scenario that challenges the common support assumption in the literature.

**Number of causal vs anti-causal variables:** Recent benchmarks [24] show that baseline methods (e.g., XGB, MLP) consistently match or outperform OOD generalization techniques (e.g., GDRO, IRM, VREX) in real-world datasets, coinciding with the prevalent causal structure  $\mathbf{X} \rightarrow Y$  in tabular data. To test if baseline superiority stems from the causal structure:  $\mathbf{X} \rightarrow Y$ , we vary the causal-to-anti-causal ratio  $\rho = |\mathbf{X}_C|/|\mathbf{X}_A| \in \{0.0, \dots, 1.0\}$  with  $|\mathbf{X}| = 50$  and  $\mathbf{X} = \mathbf{X}_C \cup \mathbf{X}_A$  (e.g.,  $\rho = 0.5$  gives  $|\mathbf{X}_C| = |\mathbf{X}_A| = 25$ ). Results (Figure D22) show baseline resilience even under the causal structures  $\mathbf{X}_C \rightarrow Y \rightarrow \mathbf{X}_A$ , though no method dominates universally (Figures D23, D24). Baselines achieve superior PMA-OOD scores [24] (fraction of maximum OOD accuracy). We evaluate: (1) ERM-based methods (XGBoost [15], LightGBM [34], MLP, ResNet [27], SAINT [76], TabTransformer, NODE [61], FT-Transformer, ExpGrad [2]); (2) OOD

**Structural Equations :**

(Hidden Confounder)  $U \sim \mathcal{N}(\mu_e, 0.5^2)$   
 $\mu_e \in \{-2, 2\}$  for ID  
 $\mu_e \in \{0, 4\}$  for OOD  
 (Input)  $X = 0.75U + \mathcal{N}(0, 1.5^2)$   
 (Outcome)  $Y = X^2 - 3U + \mathcal{N}(0, 0.5^2)$   
 (Informative)  $X_i = 0.6U + \mathcal{N}(0, 0.3^2)$

▲ Train data  
 ▼ Test data  
 ★ Predictions on train data  
 ✱ Predictions on test data

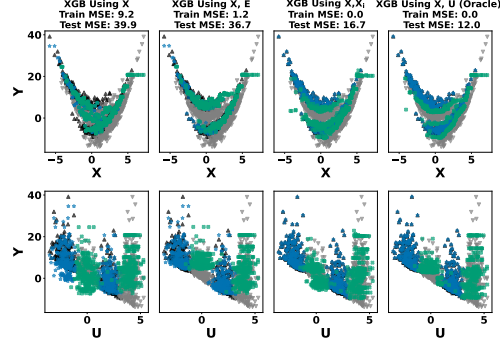


Figure D20: Performance of XGB on the low overlap non-linear setting, for sets of features: (i)  $X$  (ii)  $X$  and  $E$  (environment statistics) (iii)  $X$  and  $X_i$  (iv)  $X$  and  $U$  (Oracle).

**Structural Equations :**

(Hidden Confounder)  $U \sim \mathcal{N}(\mu_e, 0.5^2)$   
 $\mu_e \in \{-1, 0\}$  for ID  
 $\mu_e \in \{-0.5, 0.5\}$  for OOD  
 (Input)  $X = 0.75U + \mathcal{N}(0, 1.5^2)$   
 (Outcome)  $Y = X^2 - 3U + \mathcal{N}(0, 0.5^2)$   
 (Informative)  $X_i = 0.6U + \mathcal{N}(0, 0.3^2)$

▲ Train data  
 ▼ Test data  
 ★ Predictions on train data  
 ✱ Predictions on test data

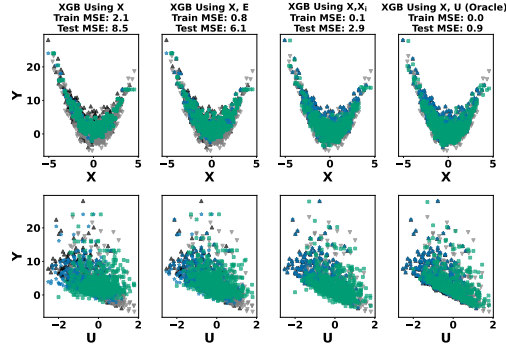


Figure D21: Performance of XGB on the low overlap non-linear setting, for sets of features: (i)  $X$  (ii)  $X$  and  $E$  (environment statistics) (iii)  $X$  and  $X_i$  (iv)  $X$  and  $U$  (Oracle).

877 generalization methods (IRM [6], IB-IRM, IB-ERM [30], CausIRL [17], DANN [3], MMD [43],  
 878 DeepCORAL [81], V-REx [38], Domain Mixup [90, 91]); and (3) Robust optimization methods  
 879 (DRO [42], GroupDRO [69], Label DRO, Adversarial Label DRO [94]).

Percentage of Max Accuracy (PMA-OOD) on Synthetic Datasets Across All ratios ( $\rho = \frac{|X_c|}{|X_t|}$ )  
 Left: Data With Linear Equations, Right: Data With Nonlinear Equations

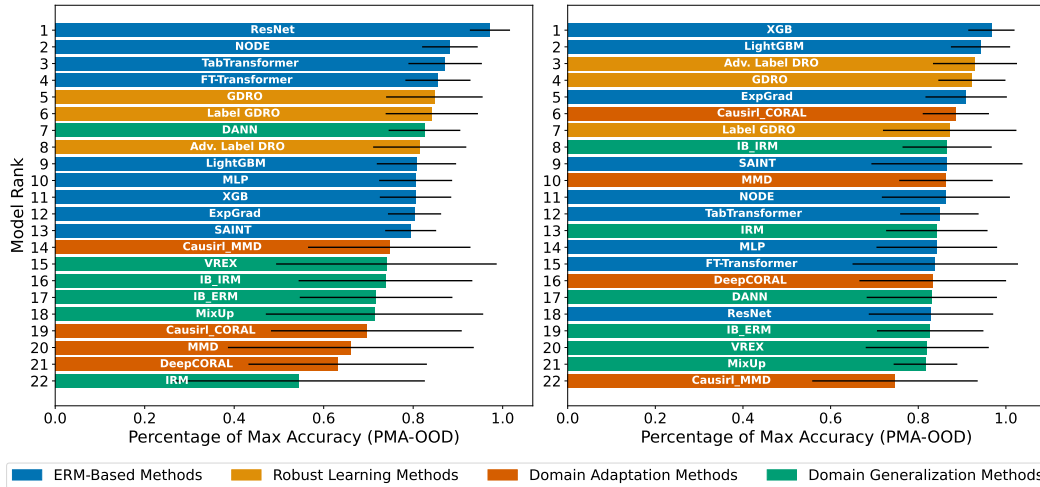


Figure D22: ERM-based methods achieve a better percentage of maximum OOD (PMA-OOD) test accuracy on the synthetic datasets, in line with the real-world results of [24].

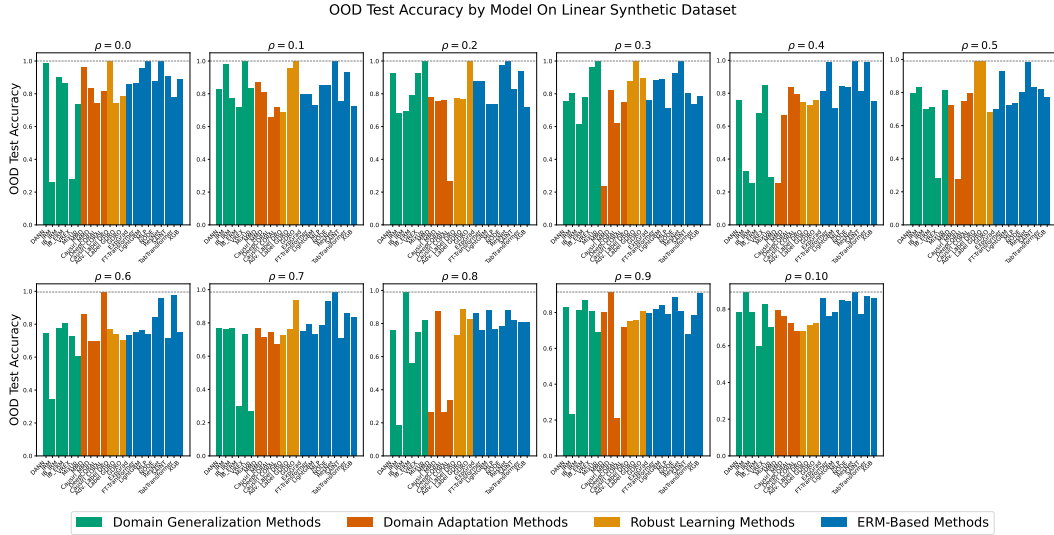


Figure D23: OOD test accuracy on the linear setting per model for different values of  $\rho$ .

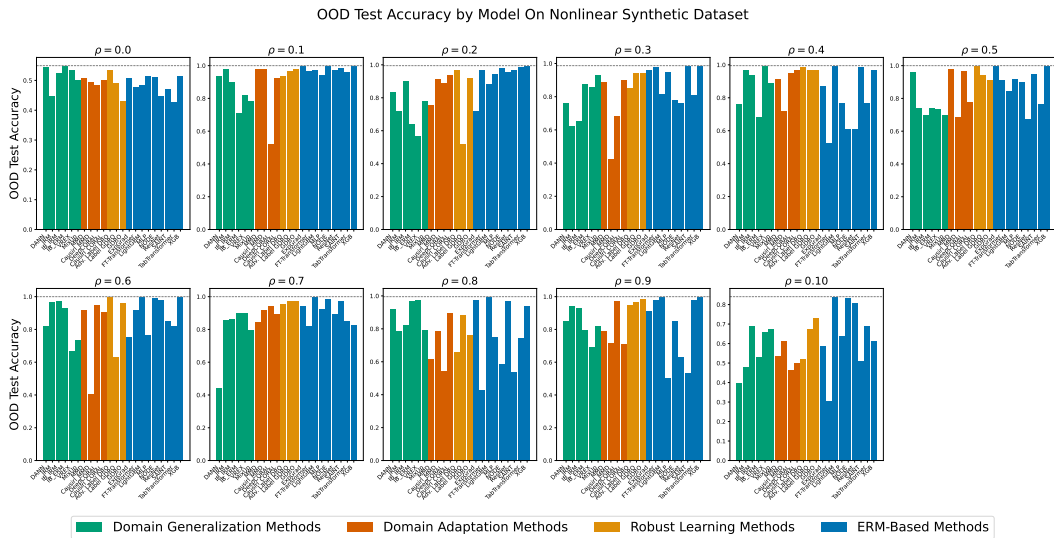


Figure D24: OOD test accuracy on the non-linear setting per model for different values of  $\rho$

## E Qualitative analysis of hidden confounding shift in real-world datasets

In this section, we qualitatively study the presence of hidden confounding shifts in real-world datasets. Tables E3-E18 present potential hidden confounders influencing various observed covariates in several real-world datasets [57]. For additional details regarding these datasets, we refer to [24, 57]. We use GPT4o [1] for this task, and we acknowledge that the list of unobserved confounding variables provided is not exhaustive. This study supports the arguments for the existence and impact of hidden confounding shifts across domains in real-world datasets.

**Prompt:** We use the prompt below to query GPT4o to get possible hidden confounders.

For a target variable, you will be given lists of causal, arguably causal, anti-causal, and spurious covariates. Also, you will be provided with environment variables such that different datasets are collected in environments induced by these environment variables. Provide possible hidden confounders that cause any of the target, causal, arguably causal, spurious, and anti-causal covariates. That is, the hidden confounders should influence the distribution of their children in different domains. Ensure hidden confounders are not a part of the given variables, and your explanation should include how hidden confounders influence their children in different environments.

Table E3: **Dataset:** Food stamps. **Target:** Food stamp reciprocity in the past year for households with child. **Environments:** regions in the United States.

| Hidden confounder                                        | Affected variables                                                                              | Reason for confounding                                                                                                                                                                           |
|----------------------------------------------------------|-------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Local economic conditions                                | Food stamp reciprocity, Age, Sex, Race, Marital status                                          | Poor economic conditions can increase food insecurity (raising food stamp usage), while also influencing population demographics due to migration, employment patterns, and household formation. |
| State or regional public assistance policies             | Food stamp reciprocity, Marital status, Number of children, Household income                    | Generous or restrictive assistance policies directly affect food stamp eligibility and indirectly influence household structure and income distribution.                                         |
| Cultural attitudes towards welfare and public assistance | Food stamp reciprocity, Marital status, Disability, Race, Ethnicity                             | Social stigma or support for welfare affects both the uptake of food stamps and social norms around marriage, disability reporting, and identity categories.                                     |
| Regional health care conditions                          | Food stamp reciprocity, Disability, Cognitive difficulty, Hearing difficulty, Vision difficulty | Regions with limited healthcare access may show higher disability prevalence and greater reliance on food stamps due to increased financial and care burdens.                                    |

Table E4: **Dataset:** Income. **Target:** Total person’s income  $\geq 56K$  for employed adults. **Environments:** regions in the United States.

| Hidden confounder                            | Affected variables                                              | Reason for confounding                                                                                                                                                |
|----------------------------------------------|-----------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Local economic conditions                    | Income, Occupation, Educational attainment, Marital status      | Stronger local economies offer better-paying jobs and education access, increasing income; they also influence occupational choices and household formation patterns. |
| State or regional tax and labor policies     | Income, Class of worker, Educational attainment, Marital status | Tax incentives and labor protections affect wages and employment types, while also shaping decisions about education and family due to financial security.            |
| Cost of living and regional affordability    | Income, Occupation, Marital status, Educational attainment      | Higher living costs necessitate higher incomes and may drive occupational or educational shifts; they also influence marriage or cohabitation decisions.              |
| Cultural norms and regional economic history | Income, Educational attainment, Occupation                      | Cultural values and historical industry presence shape education levels and job opportunities, which in turn affect income distributions.                             |

Table E5: **Dataset:** Public coverage. **Target:** Coverage of non-Medicare eligible low-income individuals. **Environments:** Disability statuses.

| Hidden confounder                                    | Affected variables                                                                                     | Reason for confounding                                                                                                                                                        |
|------------------------------------------------------|--------------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| State or regional healthcare policies                | Public health coverage, Marital status, Employment status, Income                                      | Differences in Medicaid expansion and public insurance eligibility influence health coverage; these policies also affect economic stability, employment, and family dynamics. |
| Economic conditions and poverty levels               | Public health coverage, Employment status, Income, Marital status                                      | Poorer regions tend to have lower employment rates and incomes, which increase reliance on public health coverage and influence marriage and household composition.           |
| Cultural attitudes towards disability and healthcare | Public health coverage, Disability status, Cognitive difficulty, Hearing difficulty, Vision difficulty | Stigma or support for disability and public care varies culturally, affecting both the reporting of disabilities and the likelihood of seeking or receiving public coverage.  |
| Urban vs. rural divide                               | Public health coverage, Educational attainment, Marital status, Occupation                             | Urban areas typically offer better healthcare access, education, and jobs, all of which affect coverage likelihood and socioeconomic indicators.                              |

Table E6: **Dataset:** Unemployment. **Target:** Classify whether a person is unemployed. **Environments:** Educational attainments.

| Hidden confounder                           | Affected variables                                                                                | Reason for confounding                                                                                                                                     |
|---------------------------------------------|---------------------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Local economic conditions                   | Employment status, Occupation, Marital status, Mobility status                                    | Regional economic strength affects job availability and unemployment rates; it also shapes occupation types, migration decisions, and household stability. |
| State or regional labor market policies     | Employment status, Occupation, Educational attainment, Marital status                             | Labor regulations and unemployment benefits vary by region, affecting hiring practices, education incentives, and family structures.                       |
| Cultural and social norms around employment | Employment status, Disability status, Marital status                                              | Cultural attitudes toward work and dependency influence unemployment reporting and societal roles around disability and family responsibilities.           |
| Health and disability status                | Employment status, Disability status, Cognitive difficulty, Hearing difficulty, Vision difficulty | Poor health or disabilities reduce the ability to work, directly increasing unemployment and shaping associated health-related variables.                  |

Table E7: **Dataset:** Voting. **Target:** Classify whether a person voted in the U.S. presidential election. **Environments:** United States census regions.

| Hidden confounder                                 | Affected variables                                                                     | Reason for confounding                                                                                                                         |
|---------------------------------------------------|----------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------|
| Political, family, and peer influences            | Voted in national election, Party identification, Political participation              | Social networks shape political ideology and engagement, influencing both voting likelihood and party alignment.                               |
| Media consumption habits                          | Voted in national election, Party identification, Political knowledge, Voting behavior | Media exposure affects awareness of political issues and biases, influencing party affiliation, political knowledge, and voting participation. |
| Social capital                                    | Voted in national election, Political participation, Interest in elections             | Strong community ties and civic networks increase political interest and participation, leading to higher voter turnout.                       |
| Civic education and political engagement programs | Voted in national election, Interest in elections, Political knowledge                 | Educational programs raise political awareness and civic responsibility, influencing both knowledge levels and voting decisions.               |
| Historical and cultural context                   | Voted in national election, Party identification, Interest in elections                | Historical events and regional political culture affect interest in elections and party alignment, which in turn influence voting behavior.    |

Table E8: **Dataset:** Hypertension. **Target:** Whether a person has hypertension. **Environments:** Body Mass Index (BMI) values.

| Hidden confounder                           | Affected variables                                                                                                                                | Reason for confounding                                                                                                                                                                                        |
|---------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Socioeconomic status                        | Income, Employment status, Smoking habits, Alcohol consumption, Physical activity, Healthcare access, Medical cost, High blood pressure diagnosis | Lower socioeconomic status reduces access to healthcare and healthy lifestyle options, leading to poor diet, limited activity, and delayed diagnosis, which jointly influence both BMI and hypertension risk. |
| Access to healthcare services               | Healthcare access, Medical costs, Smoking habits, Alcohol consumption, Physical activity, High blood pressure diagnosis                           | Limited healthcare access results in underdiagnosis and unmanaged hypertension, while also affecting lifestyle choices that vary with BMI, confounding the relationship between BMI and hypertension.         |
| Genetic predisposition                      | Age group, Race, Sex, Smoking habits, Diabetes, High blood pressure diagnosis                                                                     | Genetic risk factors for hypertension may co-vary with demographic attributes and influence both hypertension prevalence and BMI distribution across subpopulations.                                          |
| Psychosocial stress                         | Smoking habits, Alcohol consumption, Physical activity, High blood pressure diagnosis, Diabetes, Age group                                        | Chronic stress alters behavior (e.g., smoking, inactivity) and physiological responses, contributing to both increased BMI and elevated blood pressure, thus confounding the BMI–hypertension link.           |
| Environmental factors                       | Physical activity, Diet, Smoking habits, High blood pressure diagnosis, Diabetes, BMI category                                                    | Living environments affect access to recreational spaces, food quality, and pollution exposure, influencing both BMI and hypertension risks. These vary across BMI categories, creating confounding.          |
| Dietary habits beyond fruits and vegetables | Alcohol consumption, Smoking habits, Physical activity, High blood pressure diagnosis, BMI category                                               | High-sodium or processed-food diets raise both BMI and hypertension risk. Variation in such unmeasured dietary habits across BMI categories creates spurious associations with hypertension.                  |

Table E9: **Dataset:** College Scorecard. **Target:** Predict completion rate for first-time, full-time students at four-year institutions. **Environments:** Based on Carnegie Classifications.

| Hidden confounder                    | Affected variables                                                                                                                 | Reason for confounding                                                                                                                                                                                     |
|--------------------------------------|------------------------------------------------------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Institutional funding and re-sources | Accreditor, Control of institution, Highest degree awarded, In-state tuition, Out-of-state tuition, Cost of attendance, SAT scores | Wealthier institutions can offer better academic support, facilities, and programs, leading to higher completion rates and more selective admission profiles. This varies across Carnegie classifications. |
| Regional socio-economic factors      | Region, Poverty rate, Unemployment rate, SAT scores                                                                                | Economic conditions across regions affect affordability, student preparedness, and institutional support levels, all influencing both enrollment outcomes and graduation likelihood.                       |
| Demographic factors                  | HBCU flag, Federal loan recipient rate, Pell grant recipient rate, ACT scores, Undergraduate enrollment                            | Student demographics shape financial aid needs, academic preparation, and graduation rates. Their effect differs by institution type and selectivity under Carnegie categories.                            |
| Community support and engagement     | Distance-education flag, Federal loan recipient rate, Pell grant recipient rate, SAT scores, Undergraduate enrollment              | Supportive institutional communities improve retention and completion. Variation in engagement across institution types and student aid profiles induces confounding.                                      |
| Admission selectivity                | Admission rate, SAT (reading/math) midpoints, ACT midpoint, Undergraduate enrollment                                               | Selective admissions correlate with better-prepared students and higher completion rates, and vary with institutional prestige and classification.                                                         |
| State and local policies             | Region, Poverty rate, Unemployment rate, Cost of attendance                                                                        | Differences in education funding and public policy affect cost structures and completion outcomes, interacting with institutional classification and regional demographics.                                |

Table E10: **Dataset:** ASSISTments. **Target:** Predict whether a student solves a problem correctly on the first attempt in an online learning tool. **Environments:** Different schools.

| Hidden confounder                | Affected variables                                                                                                                                            | Reason for confounding                                                                                                                                                                                              |
|----------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Institutional teaching quality   | Hint count, Attempt count, Skill ID, Problem type, Tutor mode, Position, Type, First action, Milliseconds to first response, Overlap time, Average confidence | Variation in instructional quality and pedagogy across schools affects how effectively students engage with content, leading to differences in problem-solving strategies, response behavior, and emotional states. |
| Student motivation               | Hint count, Attempt count, Skill ID, Problem type, First action, Milliseconds to first response, Overlap time, Average confidence                             | Differences in intrinsic motivation across schools influence students' willingness to persevere, seek help, or give up quickly, affecting interaction and performance.                                              |
| Classroom environment            | Hint count, Attempt count, Tutor mode, Position, Type, First action, Average confidence                                                                       | Peer dynamics, classroom culture, and noise levels vary by school and affect how confidently and independently students solve problems.                                                                             |
| School technology infrastructure | Hint count, Attempt count, Tutor mode, Position, Type, First action, Milliseconds to first response, Overlap time, Average confidence                         | Access to reliable devices and fast internet differs by school, influencing response time, tool usage, and student experience.                                                                                      |
| Teacher-student interaction      | Hint count, Attempt count, Tutor mode, Position, Type, First action, Milliseconds to first response, Overlap time, Average confidence                         | The level of teacher guidance and feedback shapes how much support students require during problem-solving, affecting engagement and confidence.                                                                    |
| Previous academic performance    | Hint count, Attempt count, Skill ID, Problem type, First action, Average confidence                                                                           | Students' prior achievement affects how easily they solve problems, their need for assistance, and their confidence, all of which vary across schools.                                                              |

Table E11: **Dataset:** ICU. **Target:** Predict whether the patient will stay in the ICU for longer than 3 days. **Environments:** Insurance types.

| Hidden confounder                       | Affected variables                                                                                                                                                                             | Reason for confounding                                                                                                                                                                                                   |
|-----------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Socioeconomic status (SES)              | Age, Gender, Ethnicity, Height, Weight, Bicarbonate, CO <sub>2</sub> , pCO <sub>2</sub> , pO <sub>2</sub> , Lactate, Sodium, Hemoglobin, Oxygen saturation, Respiratory rate, etc.             | SES influences access to healthcare, preventive services, and overall health status. Differences in SES across insurance types lead to variability in pre-ICU health, physiological indicators, and ICU stay duration.   |
| Hospital resources and care quality     | Age, Gender, Ethnicity, Height, Weight, Bicarbonate, CO <sub>2</sub> , pCO <sub>2</sub> , pO <sub>2</sub> , Lactate, Sodium, Hemoglobin, Oxygen saturation, Respiratory rate, Heart rate, etc. | Hospital infrastructure and care standards affect monitoring, intervention speed, and clinical decisions. These factors vary by insurance coverage and influence ICU outcomes and vitals.                                |
| Comorbidities                           | Age, Gender, Ethnicity, Height, Weight, Bicarbonate, CO <sub>2</sub> , pCO <sub>2</sub> , pO <sub>2</sub> , Lactate, Sodium, Hemoglobin, Oxygen saturation, Respiratory rate, Heart rate, etc. | Presence of chronic conditions (e.g., diabetes, cardiovascular disease) affects both the need for prolonged ICU care and physiological measurements. The distribution of comorbidities differs across insurance types.   |
| Insurance-related treatment variability | Age, Gender, Ethnicity, Height, Weight, Bicarbonate, CO <sub>2</sub> , pCO <sub>2</sub> , pO <sub>2</sub> , Lactate, Sodium, Hemoglobin, Oxygen saturation, Respiratory rate, etc.             | Differences in treatment timing, intensity, and access to specialists based on insurance policies affect ICU stay duration and clinical metrics.                                                                         |
| Genetic factors                         | Age, Gender, Ethnicity, Height, Weight, Bicarbonate, CO <sub>2</sub> , pCO <sub>2</sub> , pO <sub>2</sub> , Lactate, Sodium, Hemoglobin, etc.                                                  | Inherited traits influence predisposition to organ failure, metabolic responses, and recovery trajectories. These effects are partially mediated by ethnicity and age distributions, which vary across insurance groups. |

Table E12: **Dataset:** Hospital mortality. **Target:** Classify whether an ICU patient expires in the hospital during their current visit. **Environments:** Insurance types.

| Hidden confounder                   | Affected variables                                                                                                                                                                                                                                           | Reason for confounding                                                                                                                                                                                                      |
|-------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Socioeconomic status (SES)          | Age, Gender, Ethnicity, Height, Weight, Bicarbonate, Lactate, Sodium, Hemoglobin, Oxygen saturation, Respiratory rate, Systolic blood pressure, White blood cell count, etc.                                                                                 | SES shapes access to timely and high-quality care, preventive services, and general health status. Patients with higher SES often have better insurance and outcomes, leading to confounding with mortality risk.           |
| Comorbidities                       | Age, Gender, Ethnicity, Height, Weight, Bicarbonate, CO <sub>2</sub> , pCO <sub>2</sub> , pO <sub>2</sub> , Lactate, Sodium, Hemoglobin, Oxygen saturation, Respiratory rate, Heart rate, Systolic blood pressure, etc.                                      | Pre-existing conditions such as diabetes or heart disease increase mortality risk and influence physiological features. Their prevalence differs by insurance type, creating confounding.                                   |
| Hospital resources and care quality | Age, Gender, Ethnicity, Height, Weight, Bicarbonate, CO <sub>2</sub> , pCO <sub>2</sub> , pO <sub>2</sub> , Lactate, Sodium, Hemoglobin, Oxygen saturation, Respiratory rate, Heart rate, Systolic blood pressure, White blood cell count, etc.              | Access to advanced treatments, trained staff, and timely interventions influences survival rates. These factors correlate with insurance coverage, confounding mortality outcomes.                                          |
| Genetic factors                     | Age, Gender, Ethnicity, Height, Weight, Bicarbonate, CO <sub>2</sub> , pCO <sub>2</sub> , pO <sub>2</sub> , Lactate, Sodium, Hemoglobin, Oxygen saturation, Respiratory rate, Heart rate, etc.                                                               | Genetic predispositions affect disease susceptibility and treatment responses. Variations in genetic risk factors may correlate with demographic traits across insurance types.                                             |
| Lifestyle and behavioral factors    | Age, Gender, Ethnicity, Height, Weight, Bicarbonate, CO <sub>2</sub> , pCO <sub>2</sub> , pO <sub>2</sub> , Lactate, Sodium, Hemoglobin, Oxygen saturation, Respiratory rate, Heart rate, Temperature, Systolic blood pressure, White blood cell count, etc. | Behaviors such as smoking, diet, and physical activity affect long-term health and mortality risk. These behaviors vary systematically with SES and insurance coverage, influencing both target and physiological features. |

Table E13: **Dataset:** Childhood lead. **Target:** Predict blood lead levels above CDC blood level reference value. **Environments:** Poverty-income ratios.

| Hidden confounder      | Affected variables                                                                  | Reason for confounding                                                                                                                                                                                                                                                                                                                                                                                                                                |
|------------------------|-------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Environmental Exposure | Blood lead levels, Country of birth, Age, Race and Hispanic origin                  | Environmental exposure to lead influences blood lead levels, and this varies significantly across socio-economic groups. People in lower PIR groups are more likely to live in areas with higher lead contamination, which contributes to higher blood lead levels. Furthermore, environmental factors may affect the demographic distribution (e.g., country of birth, race).                                                                        |
| Access to Healthcare   | Blood lead levels, Age, Gender, Race and Hispanic origin, Marital status, Education | Limited access to healthcare, especially in lower PIR groups, means fewer opportunities for detection and treatment of lead poisoning. This results in higher blood lead levels, with disparities also influencing demographic variables like age, gender, and education. Additionally, healthcare access varies by insurance and socio-economic status, further confounding the relationships.                                                       |
| Diet and Nutrition     | Blood lead levels, Age, Gender, Race and Hispanic origin                            | Dietary factors, such as poor nutrition in lower PIR groups, can exacerbate lead absorption. Malnutrition increases the body's susceptibility to lead poisoning, raising blood lead levels. In contrast, higher PIR groups may have better access to nutritious foods, lowering lead absorption, thus creating a confounding effect in how socio-economic status and race influence lead toxicity.                                                    |
| Housing Conditions     | Blood lead levels, Country of birth, Race and Hispanic origin, Marital status       | Older housing conditions, which are more prevalent in lower PIR groups, contribute significantly to elevated lead exposure (e.g., lead paint, poor plumbing). These living conditions directly influence blood lead levels and can also correlate with demographic factors like country of birth, race, and marital status. This introduces confounding, as socio-economic status impacts both exposure and the demographics of affected individuals. |
| Occupation             | Blood lead levels, Age, Race and Hispanic origin, Marital status, Education         | Certain occupations, which are more common among lower PIR groups, involve higher lead exposure (e.g., construction, manufacturing). Occupational lead exposure directly impacts blood lead levels and is often correlated with education, marital status, and socio-economic status. The varying prevalence of lead exposure by occupation introduces confounding, especially across different PIR groups.                                           |

Table E14: **Dataset:** Diabetes. **Target:** Predict diabetes. **Environments:** Preferred race categories.

| Hidden confounder                             | Affected variables                                                              | Reason for confounding                                                                                                                                                                                                                                                                                                                                                       |
|-----------------------------------------------|---------------------------------------------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Genetic predisposition                        | Diabetes, BMI, High blood pressure, High blood cholesterol                      | Genetic factors and family history contribute to both the onset of diabetes and comorbid conditions like obesity, hypertension, and high cholesterol. These genetic predispositions can make individuals more susceptible to diabetes, leading to confounding as they correlate with other health indicators.                                                                |
| Access to healthcare                          | Diabetes, Physical health, BMI, Healthcare coverage, Health checkups            | Limited or unequal access to healthcare, especially in marginalized racial groups, leads to disparities in diabetes diagnosis, management, and comorbidity treatment. It also influences the frequency of health checkups and access to medications, which can confound the relationship between diabetes status and other health metrics.                                   |
| Dietary habits and food availability          | Diabetes, BMI, Physical health, Alcohol consumption, Fruit and vegetable intake | Dietary habits, often shaped by socioeconomic status and local food environments, influence weight, health behaviors (such as alcohol consumption), and diabetes risk. People in lower socioeconomic strata may have limited access to healthy food options, leading to higher BMI and increased diabetes risk, creating confounding effects on health outcomes.             |
| Psychosocial stress and mental health factors | Diabetes, Mental health, Physical health, BMI, Physical activity, Doctor visits | Chronic stress and mental health issues, often higher in marginalized groups, contribute to diabetes development and complicate its management. These factors also affect physical health (e.g., weight gain due to stress) and health-seeking behaviors (e.g., fewer doctor visits), leading to confounding by influencing both diabetes risk and its associated variables. |
| Socioeconomic status beyond income            | Diabetes, Income, Physical health, Healthcare coverage, Education level         | Socioeconomic factors, such as occupation, education, and neighborhood wealth, influence access to healthcare, nutrition, and overall health behaviors. These factors can confound the relationships between diabetes and other socio-economic variables like income and education, as they shape opportunities for prevention and treatment.                                |

Table E15: **Dataset:** Sepsis. **Target:** Predict, from a set of fine-grained ICU data, whether a patient will experience sepsis onset within the next 6 hours. **Environments:** Lengths of ICU stay.

| Hidden confounder                                    | Affected variables                                                                                                   | Reason for confounding                                                                                                                                                                                                                                                                                                                                                                                                     |
|------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Infection prevalence in ICU                          | SepsisLabel, Temperature (Temp), Leukocyte count (WBC), Heart rate (HR), Blood urea nitrogen (BUN)                   | Higher infection rates in certain ICU units can lead to a higher probability of sepsis onset (SepsisLabel). These infection rates influence biomarkers such as WBC, HR, and BUN, creating confounding because the unit's infection environment affects both the likelihood of sepsis and the observed clinical measures.                                                                                                   |
| Quality of ICU Care                                  | SepsisLabel, Fibrinogen concentration (Fibrinogen), Leukocyte count (WBC), Platelet count (Platelets)                | Higher-quality care in certain ICUs may lead to earlier identification and treatment of sepsis, resulting in more accurate SepsisLabel predictions. Additionally, better care could affect biomarkers like fibrinogen, WBC, and platelets, which are critical in sepsis detection and progression, thereby confounding the relationships between these variables and the outcome.                                          |
| Severity of underlying conditions                    | SepsisLabel, Blood urea nitrogen (BUN), Creatinine, Lactate, Calcium                                                 | Patients with severe chronic conditions (e.g., kidney disease, cardiovascular issues) are at a higher risk of sepsis and may show abnormal levels in biomarkers like BUN, creatinine, lactate, and calcium. These underlying conditions contribute to the SepsisLabel outcome and confound the relationship between the biomarkers and the likelihood of sepsis, varying across ICU units depending on patient population. |
| Patient's socio-economic status                      | Age (Age), Gender (Gender), Leukocyte count (WBC), Fibrinogen concentration (Fibrinogen), Platelet count (Platelets) | Socio-economic factors, such as access to healthcare, can influence both the likelihood of sepsis and the observed clinical biomarkers. For example, patients from lower socio-economic backgrounds may have delayed hospitalizations or inadequate care, which affects both SepsisLabel and the progression of sepsis as indicated by WBC, fibrinogen, and platelet levels.                                               |
| Hospital-specific protocols and treatment guidelines | SepsisLabel, Lactate, Glucose, Creatinine                                                                            | Differences in hospital protocols for sepsis treatment, such as timing of interventions and choice of sepsis bundles, can affect both the SepsisLabel and biomarkers like lactate, glucose, and creatinine. These protocols lead to variability in how sepsis is diagnosed and treated across different hospitals and ICU units, confounding the relationship between biomarkers and sepsis outcomes.                      |

Table E16: **Dataset:** Hospital readmission. **Target:** Predict whether a diabetic patient is readmitted to the hospital within 30 days of their initial release. **Environments:** Admission sources.

| Hidden confounder                 | Affected variables                                                                                                                                                    | Reason for confounding                                                                                                                                                                                                                                                                                                                                                                               |
|-----------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Socio-economic status (SES)       | Race, Gender, Age, Payer code, Medical specialty, Number of outpatient visits, Number of emergency visits, Number of inpatient visits, Diabetes medication prescribed | SES influences access to healthcare, patient demographics, and chronic disease rates. It can also determine the type of care received based on admission source (e.g., emergency department vs. outpatient settings). Differences in healthcare access, such as availability of medications, may impact readmission rates and associated variables like outpatient visits and prescribed medication. |
| Severity of illness               | Primary diagnosis, Secondary diagnosis, Number of diagnoses, Discharge type, Medication changes (e.g., Insulin, Glipizide)                                            | More severe illness increases the likelihood of readmission and influences the complexity of diagnoses and the treatments administered. The severity of illness may differ based on the admission source (e.g., emergency versus outpatient), impacting the number and type of diagnoses and treatments prescribed at discharge, affecting readmission likelihood.                                   |
| Access to healthcare resources    | Time in hospital, Discharge disposition, Number of procedures, Number of medications, Number of lab tests                                                             | Access to healthcare resources (e.g., time in hospital, availability of procedures and medications) influences treatment decisions and outcomes. Different admission sources may have varying levels of available resources, leading to different lengths of stay, the types of procedures performed, and overall treatment quality, which can affect the likelihood of readmission.                 |
| Patient's adherence to medication | Change in medications, Diabetes medication prescribed, Number of outpatient visits, Number of emergency visits                                                        | Adherence to prescribed medications is often influenced by socio-economic status, which can vary depending on admission source. Non-adherence may lead to medication changes, affecting diabetes control and subsequent readmission risk. The number of outpatient and emergency visits can also reflect how well a patient manages their diabetes and the likelihood of complications.              |
| Hospital-specific protocols       | Time in hospital, Number of diagnoses, Discharge disposition, Readmitted                                                                                              | Different hospitals and healthcare systems implement various protocols for discharge planning and readmission prevention, which can affect readmission rates. These protocols may vary by admission source, where patients admitted via the emergency department may receive different follow-up instructions and care than those admitted through other channels.                                   |

Table E17: **Dataset:** MEPS. **Target:** Measure of health care utilization. **Environments:** Insurance types.

| Hidden confounder                  | Affected variables                                                                                                                                       | Reason for confounding                                                                                                                                                                                                                                                                                                                                                                                                    |
|------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Socioeconomic status (SES)         | Years of education, Employment status, Hourly wage, Paid sick leave, Paid leave to visit doctor, Family size, Insurance coverage, Healthcare utilization | SES can influence access to healthcare services, insurance coverage, and the ability to use medical services. Insurance types often correlate with SES levels, where individuals with lower SES may be more likely to be insured by government programs (e.g., Medicaid), which in turn impacts healthcare utilization patterns across different SES groups.                                                              |
| Healthcare access                  | Paid sick leave, Insurance coverage, Employer offers health insurance, Healthcare utilization                                                            | Limited access to healthcare, such as lack of insurance or paid sick leave, directly impacts healthcare utilization. The type of insurance a person holds is often tied to access to various medical services. The level of coverage and accessibility differs across insurance types, influencing healthcare behaviors such as whether a patient can afford and utilize healthcare services.                             |
| Health behaviors                   | Perceived health status, Asthma medications, Limitations in physical functioning, Healthcare utilization                                                 | Lifestyle factors like smoking, alcohol consumption, and physical activity can directly affect health status and healthcare needs. Health behaviors differ across groups with different insurance types, and these behaviors contribute to healthcare utilization. Insurance coverage can also be influenced by perceived health status, which varies across insured groups, affecting utilization of medical services.   |
| Chronic health conditions          | Asthma medications, Perceived health status, Limitations in physical functioning, Healthcare utilization                                                 | Chronic conditions like asthma or diabetes increase the need for healthcare services, leading to higher healthcare utilization. People with chronic conditions are more likely to be covered by Medicare or Medicaid, which influences the type of insurance they hold and impacts their healthcare utilization and associated features (e.g., medications and physical limitations).                                     |
| Regional healthcare infrastructure | Region, Family size, Healthcare utilization, Paid sick leave, Paid leave to visit doctor, Employment status                                              | The quality and availability of healthcare infrastructure vary across regions, which can impact healthcare utilization. In underserved regions, people may be more reliant on public insurance options like Medicaid, affecting healthcare access and behaviors. The regional differences in healthcare systems can lead to disparities in access to medical services, insurance coverage, and utilization of healthcare. |

Table E18: **Dataset:** Poverty. **Target:** Predict household income-to-poverty ratio. **Environments:** Citizenship statuses.

| Hidden confounder                          | Affected variables                                                                                                                               | Reason for confounding                                                                                                                                                                                                                                                                                                                                            |
|--------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Social capital                             | Income, Unemployment compensation, Disability benefits, Family size, Housing conditions, Household income-to-poverty ratio                       | Stronger social support networks can improve access to income, benefits, and resources like unemployment or disability compensation. Social capital can vary based on citizenship status, which in turn influences income, housing conditions, and eligibility for benefits, confounding the relationship between household income and poverty ratios.            |
| Workplace discrimination or bias           | Household income, Worker's compensation, Unemployment compensation, Pension, Family income, Disability benefits                                  | Discrimination in the workplace can limit opportunities for higher wages, pensions, or disability benefits, especially for marginalized groups (e.g., racial minorities, non-citizens). This bias varies by citizenship status and leads to biased associations between income, benefits, and poverty levels.                                                     |
| Healthcare access                          | Household income, Health insurance premiums, Medical expenses, Disability benefits, Family size, Health conditions, Medicaid/Medicare assistance | Limited access to healthcare, particularly for non-citizens, can lead to higher medical expenses and greater reliance on public health assistance. Citizenship status directly impacts eligibility for public healthcare programs, confounding relationships between medical expenses and income, and affecting poverty ratios.                                   |
| Access to education and skills development | Educational attainment, Household income, Employment status, Unemployment compensation, Income from assistance, Savings                          | Disparities in educational opportunities, often linked to citizenship status, influence income potential and eligibility for government assistance. These disparities, in turn, affect the household income-to-poverty ratio and employment outcomes, creating confounding relationships between education and income.                                            |
| Cultural factors                           | Family size, Household income, Savings, Disability benefits, Healthcare utilization, Income assistance, Living arrangements                      | Cultural factors influence financial management, family support, and the use of social programs. These factors can vary across citizenship statuses, leading to differences in how household income and assistance are distributed, and thus confounding the relationship between income, benefits, and financial behavior.                                       |
| Housing market and rent conditions         | Housing ownership, Housing costs, Family size, Household income, Rent payments, Social security benefits, Medical aid                            | Housing market conditions, especially rent disparities, can create financial strain and impact the household income-to-poverty ratio. Citizenship status often influences eligibility for housing assistance and rent subsidies, which confounds the relationship between housing costs and income, particularly in areas with significant immigrant populations. |